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Semiconductors | North America

Disti Survey 2Q'26: Broadly Holding Up

Disti responses remain constructive after the sharp 1Q step-up. We expect strong JunQ prints, with leading indicators broadly holding into SepQ.

AlphaWise α

Last quarter marked a clear turn in the cycle, and the 2Q26 survey suggests that strength is carrying into SepQ, albeit at a more measured pace. Expectations remain healthy after seasonality, with no respondents expecting a decline in Analog or MCU. Industrial held up, while Auto was mixed. The recovery remains healthy, though the rate of improvement has moderated after the sharp 1Q acceleration.

Our AlphaWise analysis points to a similarly uneven but constructive setup. Analog and MCU are shipping above demand, while Disti momentum eased as inventory-building intentions moderated. We view the deceleration as a more measured continuation, rather than a reversal, as momentum remains positive and the moderation has not been accompanied by weaker pricing or forward expectations. The recovery continues to look demand-led, with selective replenishment and supply tightness rather than broad restocking or shortages at this stage.

What does this mean for earnings? We expect strong JunQ prints, similar to last quarter, and raise our JunQ/SepQ estimates for **TXN**, with our updated estimates implying above-seasonal growth in both quarters, supported by Analog/Industrial strength, data center growth and firm pricing, though this does not change our UW view. We leave other estimates unchanged for now, with TXN the first test of whether these trends are translating into revenue and gross margin. **ADI** has the clearest Analog/Industrial read-through, **ON** is best exposed to selective power tightness, **NXP** could benefit if Auto is better than feared and **MCHP** remains a later-2H/4Q MCU/pricing story.

Exhibit 1: 2Q26 Disti Survey Scorecard: Constructive; 1Q Strength Broadly Holds

2Q26 Disti Survey: Healthy & Holding Up			
Section	**Signal**	**Q/Q Trend | Reading**	**%ile Rank**
Growth Expectations	↗ HEALTHY, PROTECTED DOWNSIDE	Analog ↑ MCU ↓ Power (new)	*Q/Q: 96th %ile | S: 90th %ile* *Q/Q: 76th %ile | S: 92nd %ile*
End Markets	↗ SELECTIVE UPSIDE	AI/DC (New) Industrial → Auto ↓	*93rd %ile* *46th %ile*
Inventory/ Channel	→ NORMALIZING	Inventory Builds ↓ Double orders ↑	*80th %ile* *82nd %ile*
Supply& Pricing	→ FIRM PRICING MIXED SUPPLY	Analog pricing → MCU pricing → Constraints → Lead times ↓	*89th %ile* *88th %ile* *53rd %ile* *57th %ile*

Source: AlphaWise, SIA, Morgan Stanley Research.

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WHAT'S CHANGED

	From	To
Texas Instruments (TXN.O) Price Target	$221.00	**$230.00**

Cycle Analysis: Recovery Progressing, but Still Below Prior Pace

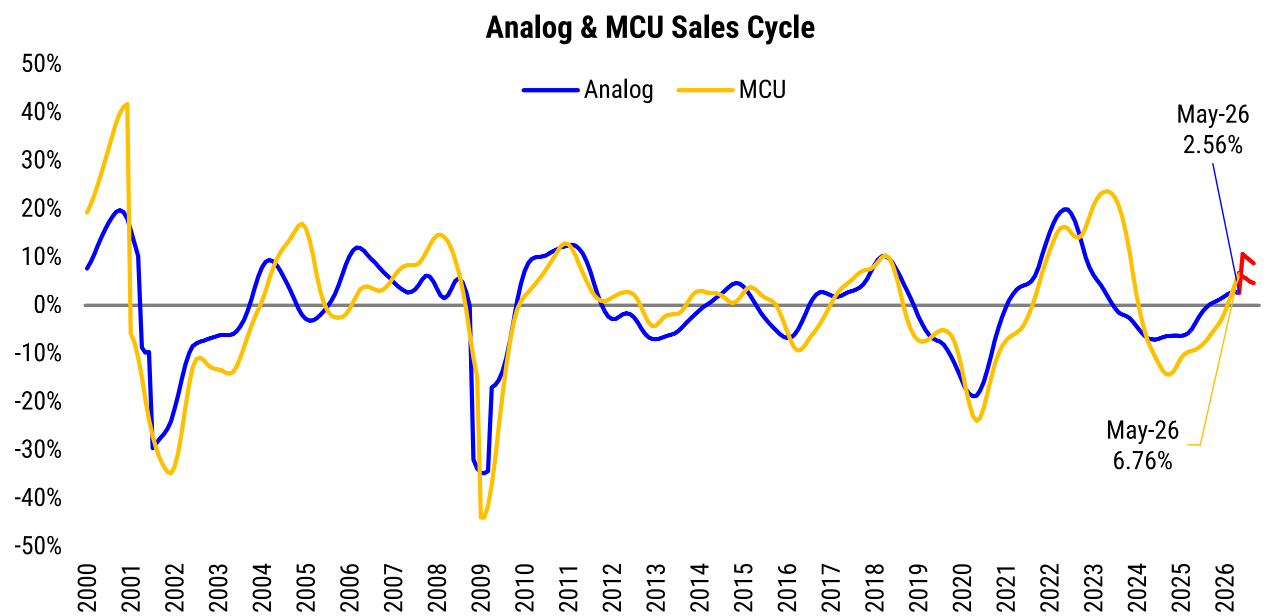
Analog, MCU and IGBT/Power have all moved to shipping above demand, but the broader recovery remains gradual. The sharp 1Q acceleration was meaningful, while the latest moderation reflects a slower rate of improvement rather than a return to contraction.

Where Are We in the Cycle?

Recovery progressing, but still slower than most prior cycles. Analog, MCU and IGBT/Power have all moved to shipping above demand, but the broader recovery remains gradual. The sharp 1Q acceleration was meaningful, while the latest moderation reflects a slower rate of improvement rather than a return to contraction.

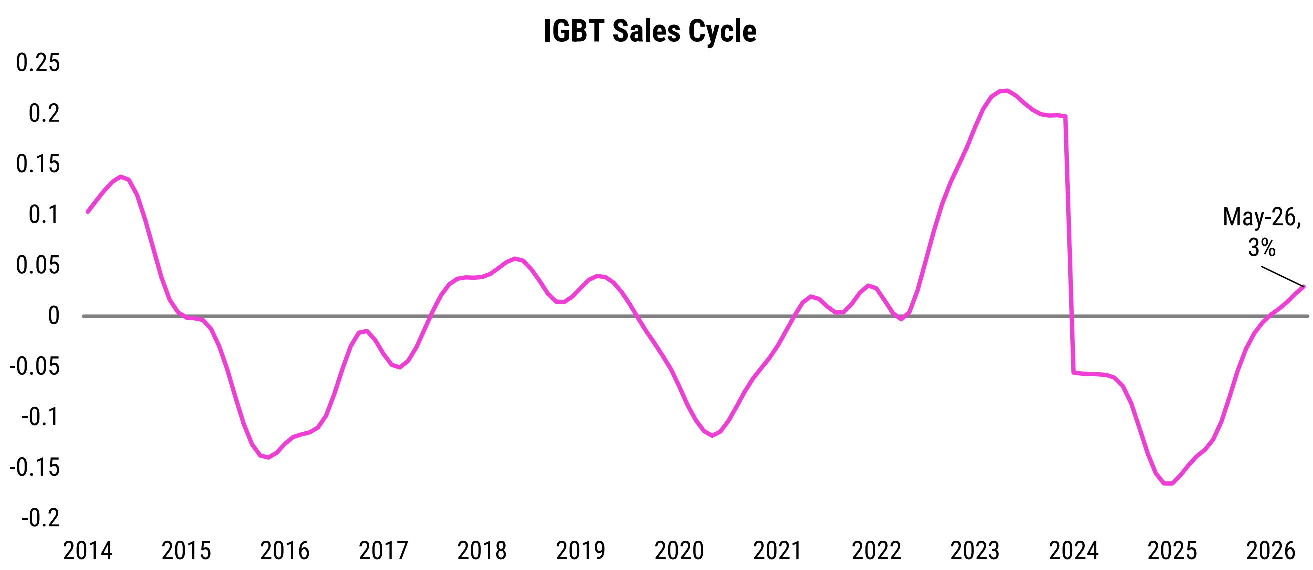
- **Analog:** bottomed in June 2024 at 7.1% below demand. As of May 2026, Analog is shipping 2.6% above demand, representing a 9.7ppt improvement over 23 months. That remains slower than the closest comparable historical recoveries, but the cycle continues to advance.
- **MCU:** bottomed later than Analog and moved above demand more recently. As of May 2026, MCU is shipping 6.8% above demand. The recovery has accelerated meaningfully since 1Q, but remains earlier and more uneven than Analog.
- **IGBT/Power:** moved from shipping ~1% below demand in February to 3% above demand in May. Power is therefore also moving into modest overshipping, supported by stronger demand around AI/data center and selected high-voltage applications.

Exhibit 2: Analog & MCU Sales Cycle - both now shipping above demand



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 3: IGBT Sales cycle: now marginally shipping above demand

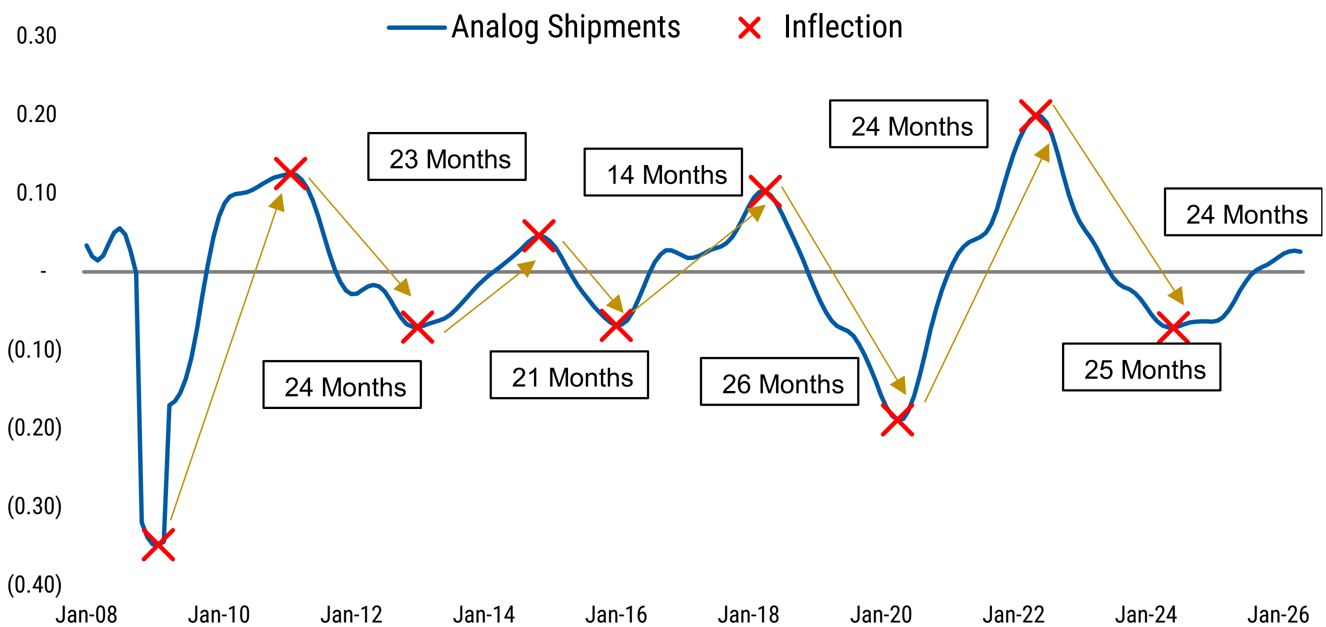


Source: AlphaWise, SIA, Morgan Stanley Research.

Analog continues to recover, but the pace remains slower than in most prior cycles. Since bottoming in June 2024, the cycle indicator has improved 9.7ppts to +2.6% through May 2026. That trails the 11.5ppt and 13.7ppt improvements seen at the same point following the January 2013 and January 2016 troughs, respectively, while the recovery following April 2020 was considerably faster. In our view, the slower path leaves further room for the cycle to run, even as near-term momentum becomes more measured.

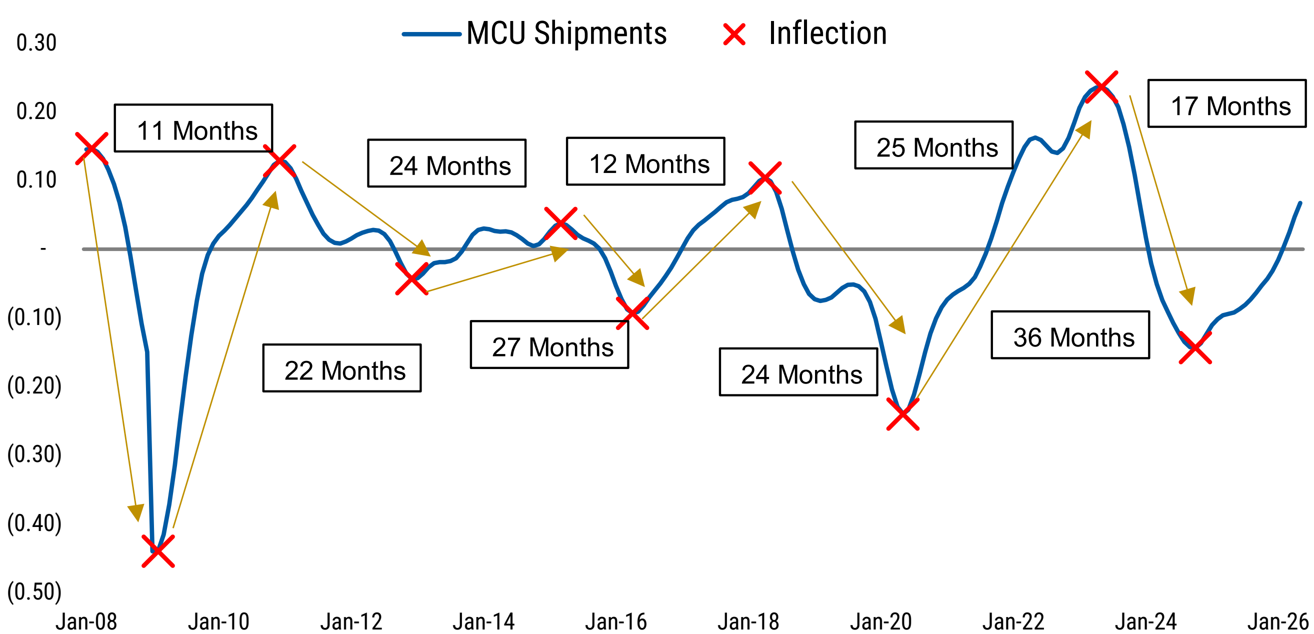
The slow pace was evident early in the recovery. During the first seven months after the June 2024 trough, Analog improved just 0.8ppt, versus a 12.6ppt average across the prior upcycles shown. The closest comparison was January-August 2013 at +2.3ppts, while the 2009 and 2020 recoveries were much stronger at +27.6ppts and +14.2ppts, respectively. These figures measure the pace of improvement over a fixed period, while the chart below measures the duration between cycle inflection points.

Exhibit 4: Analog cycles have typically lasted ~22-26 months from trough to peak or peak to trough



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 5: MCU cycle duration has been more variable, with pre-2020 cycle legs typically lasting ~22-27 months



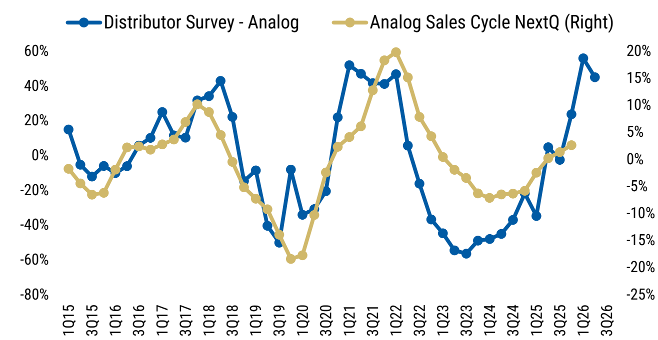
Source: AlphaWise, SIA, Morgan Stanley Research.

Taken together, Analog, MCU and IGBT/Power are all now shipping modestly above demand. That suggests the initial catch-up phase has progressed, but not that the cycle is mature: the recovery remains slower than most prior cycles, broad industry revenues remain below prior peaks and pricing continues to hold. We view the 2Q moderation as a slower continuation of the upcycle, not a rollover.

Disti Momentum - Moderating After the 1Q Jump

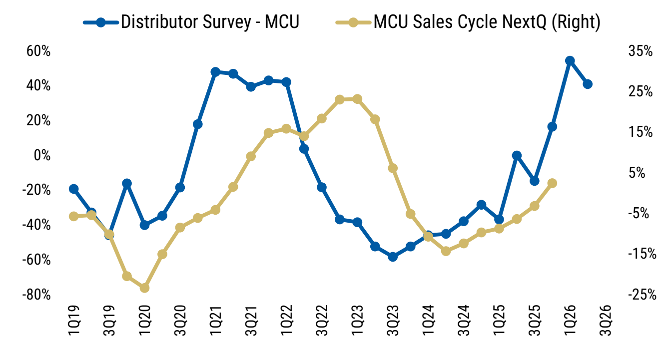
The moderation in Disti momentum is consistent with inventory-building intentions easing after the sharp 1Q increase. Because the Disti tracker is based on the spread between respondents building and depleting inventory, the 1Q jump moved ahead of the shipment recovery. Some deceleration is therefore natural as shipments catch up and inventory intentions become less aggressive. The tracker remains positive and well above the levels seen through most of the current recovery.

Exhibit 6: Analog Disti momentum moderated after the sharp 1Q increase...



Source: AlphaWise, SIA, Morgan Stanley Research

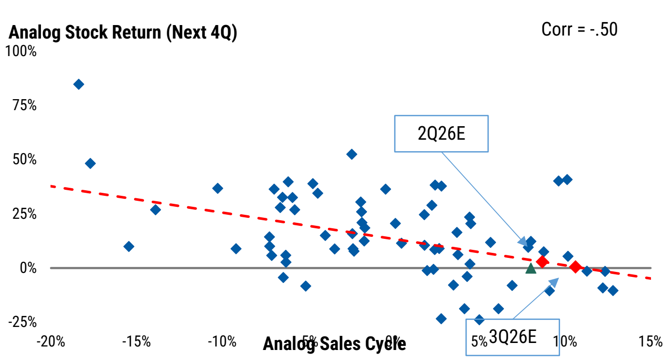
Exhibit 7: ...while MCU momentum also eased, though underlying trends remain positive



Source: AlphaWise, SIA, Morgan Stanley Research

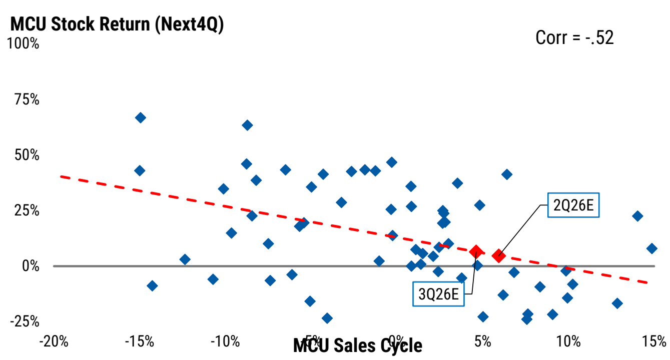
The return analysis adds an important distinction. Forward returns have historically moderated as the sales cycle moves further into overshipping, suggesting that broad cycle beta may become less powerful as the recovery advances. However, Analog and MCU remain only modestly above demand and broader industry revenues remain below prior peaks. We view this as an argument for greater stock selection, not a negative sector call.

Exhibit 8: Forward analog returns historically moderate as the sales cycle advances...



Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 9: ...with a similar relationship for MCU



Source: AlphaWise, SIA, Morgan Stanley Research

Disti Survey 2Q'26: 1Q Strength Broadly Holds as Momentum Moderates

The 1Q step-up is broadly holding. SepQ demand expectations remain healthy after accounting for seasonality, pricing remains firm and inventory-building intentions are normalizing without renewed digestion. Industrial remains constructive, Auto is more mixed and supply tightness remains selective.

AlphaWise Analysis: Healthy & Holding Up

Our 2Q26 Distributor Survey points to healthy SepQ demand across Analog/MCU. Analog sequential growth expectations increased to 75% from 71%, while MCU moderated to 54% from 64%, with no respondents expecting a decline in either product line. Given typically stronger SepQ seasonality, the more useful signal is that 95% of respondents expect demand to be in-line with or above seasonal levels. Above-seasonal expectations eased to 50% for Analog and 46% for MCU from the very strong 1Q readings, but remain elevated.

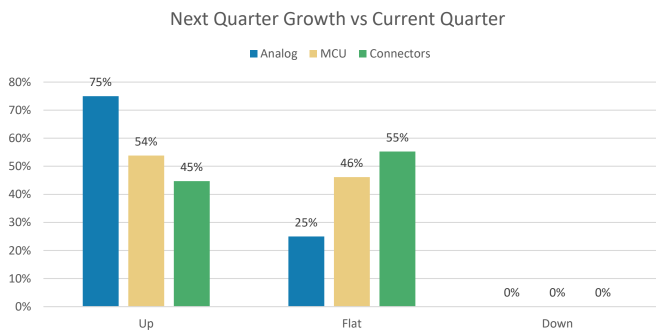
The broader scorecard tells a similar story. Analog demand and pricing remain the strongest signals, Industrial held up and inventory-building intentions eased without renewed inventory digestion. Supply signals remain selective, with pockets of tightening around MCU/power and AI/data-center applications rather than a broad shortage cycle.

Exhibit 10: 2Q26 Disti Survey Scorecard: Constructive; 1Q Strength Broadly Holds

2Q26 Disti Survey: Healthy & Holding Up			
Section	**Signal**	**Q/Q Trend | Current Reading**	**Survey Quotes**
Growth Expectations	↗ HEALTHY, PROTECTED DOWNSIDE	Analog ↑ Growth: 75% q/q (+4pp); 96th %ile rank Seasonal: 95% ≥ seasonal; 90th %ile rank MCU ↓ Growth: 54% q/q (-10pp); 76th %ile rank Power (new) Seasonal: 95% ≥ seasonal; 92nd %ile rank Demand: 63% higher	Analog: *"Higher demand...driven by AI, Datacenter, and Aerospace."* MCU: *"MCU constrained but demand still strong..."* Power: *"MCU, power anything going into AI servers."*
End Markets	↗ SELECTIVE UPSIDE	AI/DC (New) 63% higher demand Industrial → 40% net positive (+2pp); 93rd %ile rank Auto ↓ 13% net positive (-11pp); 46th %ile rank	AI/DC: *"Strong AI demand...affecting surrounding components."* Industrial: *"Data center, industrial connectors."* Auto: *"Industrial, data center, auto markets."*
Inventory/ Channel	→ NORMALIZING	Inventory ↓ 50% build (-21pp); 80th %ile rank Double orders ↑ 38% (+11pp)	Restocking: *"Customers buying additional units due to supply fears."* Order behavior: *"CMs and OEMs...ordering the same part."* Urgency: *"Front loading demand...pressure to the expediting activities."*
Supply& Pricing	→ FIRM PRICING MIXED SUPPLY	Analog pricing → 63% stronger (-2pp); 89th %ile rank MCU pricing → 59% stronger (+1pp) Constraints → 68% constrained (flat); 53rd %ile rank Lead times ↓ 53% rising (-12pp)	Analog: *"All pricing is going up."* MCU: *"All vendors are increasing pricing."* Power: *"AI boost...pushing lead times and prices up."*

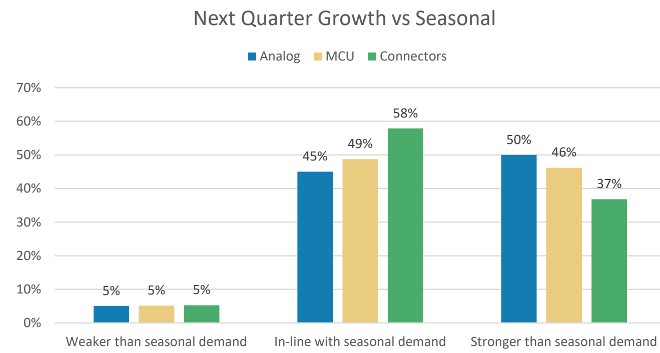
Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 11: Disti results point to positive sequential growth for the SepQ...



Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 12: ...while seasonally adjusted expectations remain healthy across Analog/MCU.



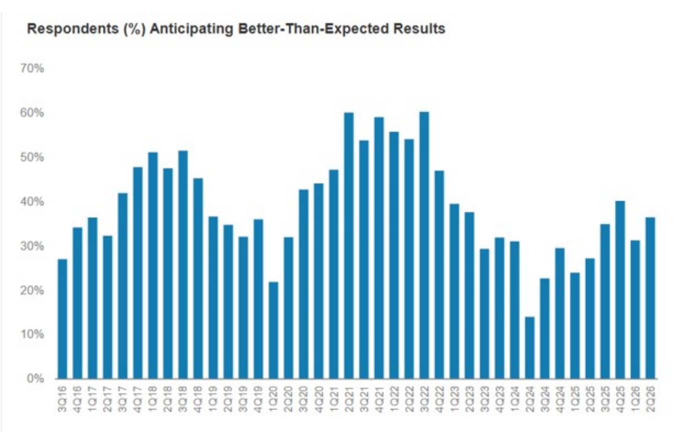
Source: AlphaWise, SIA, Morgan Stanley Research

1) Industrial Demand Holds Up; Auto Remains Mixed

Industrial remains the clearest area of strength outside data center. Stronger-than-seasonal expectations remained ~flat q/q at 47%, while the latest Multi-Industry Distributor Survey also improved, with better-than-expected results rising to 36% from 31% and FY26 volume expectations increasing to 2.8% from 2.3%. Company commentary continues to characterize the recovery as underlying demand rather than inventory build.

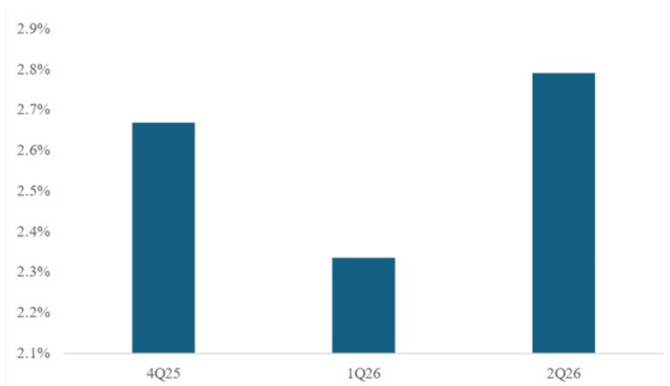
Auto is less clean. Weaker-than-seasonal responses increased to 15% from 6%, even as company commentary suggests inventory digestion is largely complete and suppliers are shipping closer to natural demand. The contrast is that Industrial is already showing a broader demand recovery, while Auto has not yet entered a broad replenishment cycle. We remain constructive on Industrial and selective in Auto.

Exhibit 13: Multi-Industry distributors continued to report better-than-expected results in 2Q26...



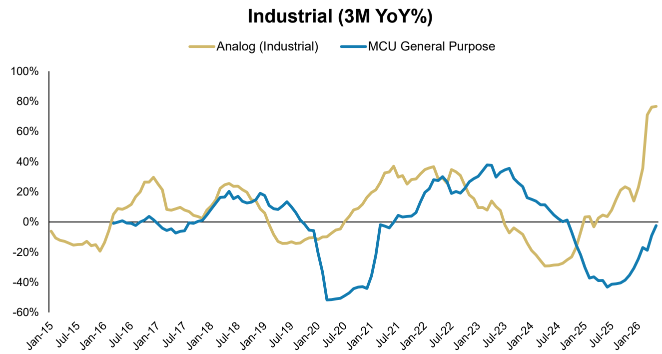
Source: AlphaWise, Morgan Stanley Research

Exhibit 14: ...while FY26 volume expectations increased



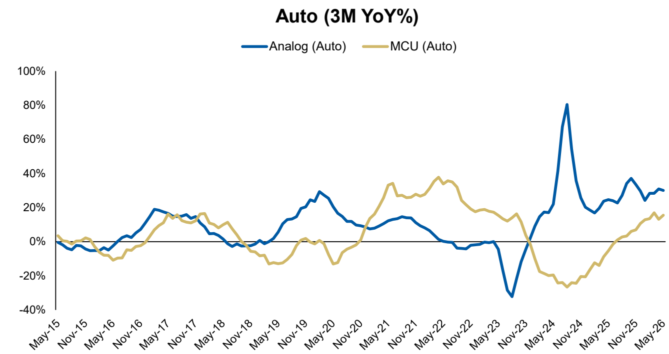
Source: AlphaWise, Morgan Stanley Research

Exhibit 15: Industrial growth has accelerated sharply...



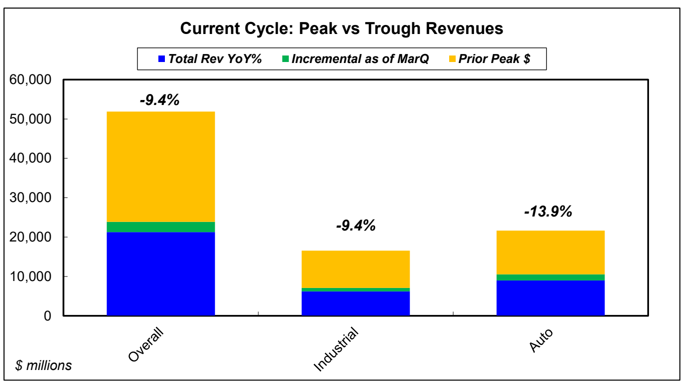
Source: SIA, Morgan Stanley Research

Exhibit 16: ...while Auto growth remained more mixed



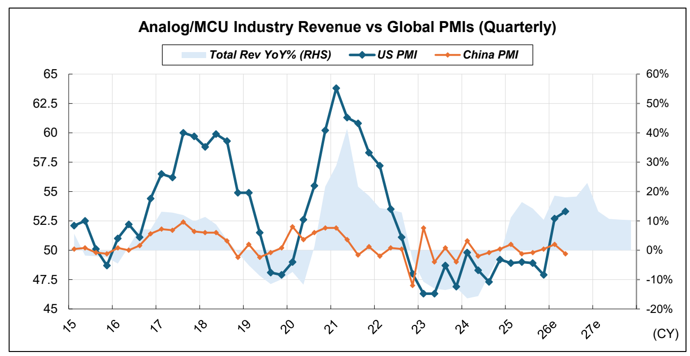
Source: SIA, Morgan Stanley Research

Exhibit 17: The industry remains below prior peaks, though Industrial improved sequentially...



Note: As of 1Q26 earnings. Source: Company data, VisibleAlpha, Morgan Stanley Research

Exhibit 18: ....with broadly improving PMIs providing further support



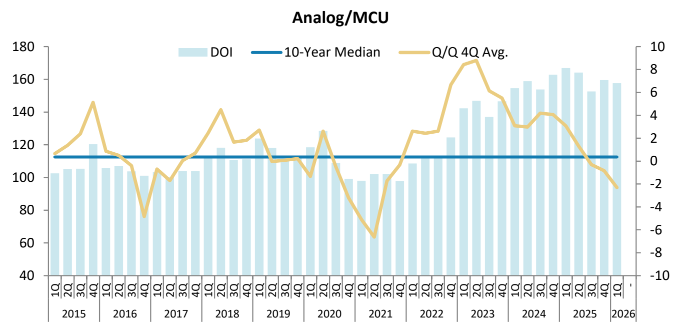
Note: Estimates are consensus. Source: Company data, VisibleAlpha, Morgan Stanley Research

2) Inventory Builds Moderate; Broad Restocking Still Absent

Inventory-building intentions moderated after the sharp 1Q increase. Overall builders declined to 50% from 71%, while depletion increased to 30% from 18%. However, builds greater than 5% remained ~flat for Analog and MCU, suggesting less aggressive inventory accumulation rather than a renewed correction.

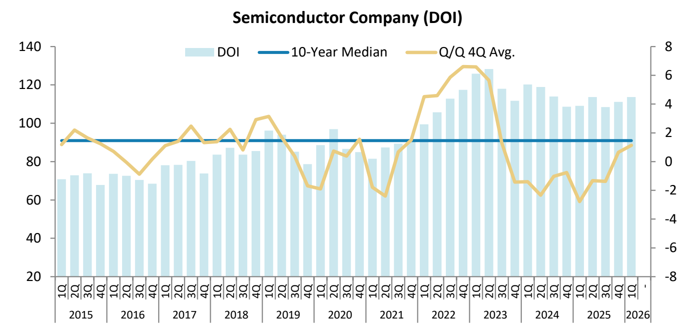
Distributor inventory declined for a fourth consecutive quarter and producer builds remained below seasonality, while customer DOI increased ~in-line with normal seasonality. Absolute inventory remains elevated, but the data do not indicate that broad inventory digestion has restarted. We continue to see selective replenishment and channel discipline rather than a broad 2H restocking cycle.

Exhibit 19: Analog/MCU inventory remains elevated, although DOI momentum has moderated since 2Q23...



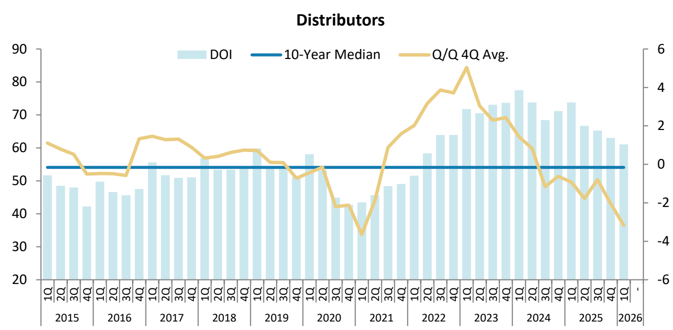
Source: FactSet, Morgan Stanley Research

Exhibit 20: ...while overall semi company DOI momentum has been increasing...



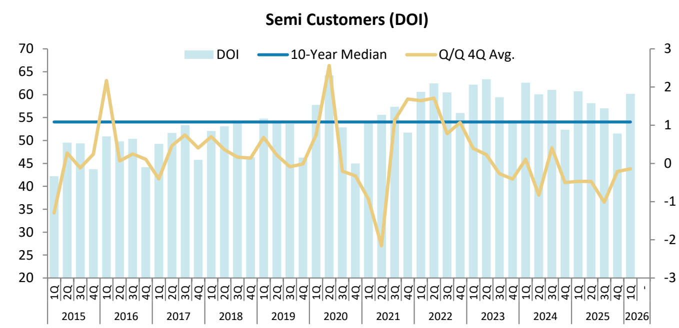
Source: FactSet, Morgan Stanley Research

Exhibit 21: ...and distributor inventory has declined for four consecutive quarters...



Source: FactSet, Morgan Stanley Research

Exhibit 22: ...while customer DOI increased, restocking ~in-line with 1Q seasonality.



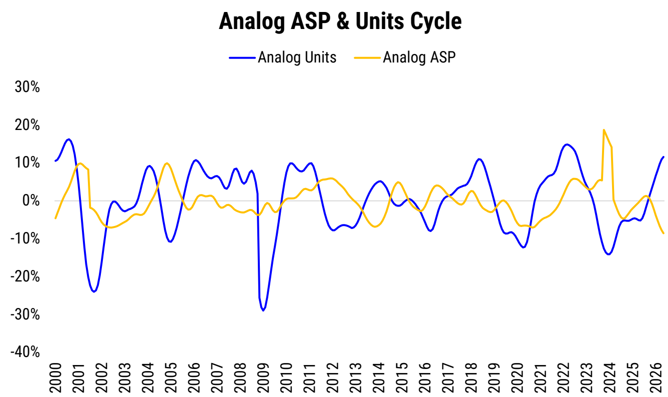
Source: FactSet, Morgan Stanley Research

3) Pricing Holds; Tightness Remains Selective

Pricing remains the strongest carryover from the 1Q survey. Stronger-than-usual Analog pricing held at 63% versus 65% in 1Q, while stronger MCU pricing was 59% versus 58%. No respondents reported weaker pricing. That matters because the moderation in Disti momentum has not been accompanied by the pricing deterioration that would make the deceleration more concerning.

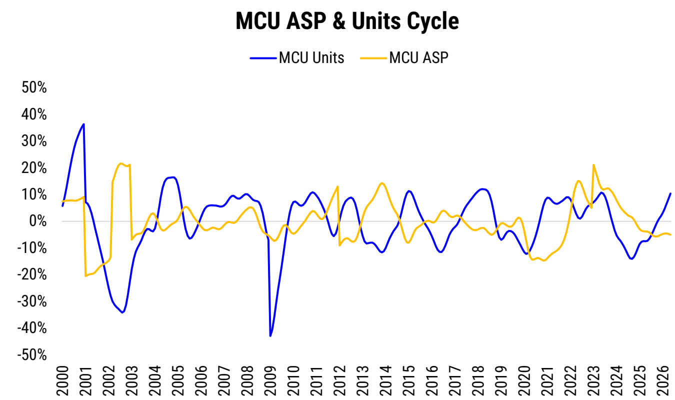
Supply signals were more mixed. Overall constraints remained at 68% and respondents reporting rising lead times declined to 53% from 65%. At the same time, Analog expedites increased, MCU constraints rose to 44% from 39% and MCU double ordering increased. Selected power lead times also tightened around AI/data-center demand. We see firm pricing and pockets of tightness, not a broad shortage cycle.

Exhibit 23 Analog unit momentum is improving as ASPs stabilize...



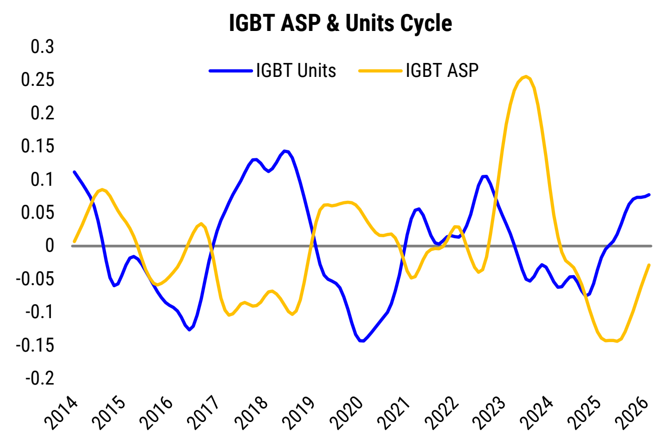
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 24 ...with MCU units recovering and ASP pressure easing...



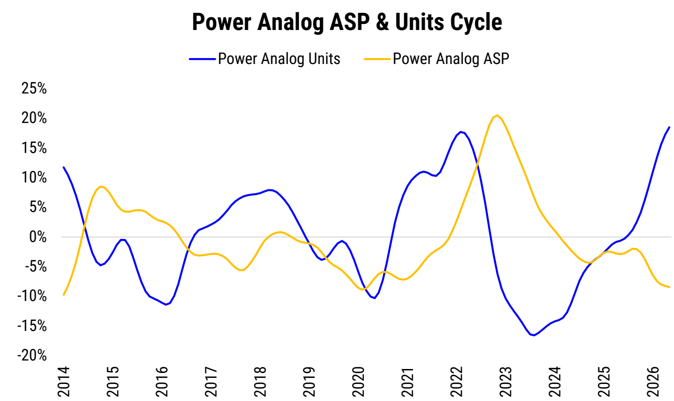
Source: AlphaWise, SIA, Morgan Stanley Research

Exhibit 25 ...while IGBT units continue to recover as ASPs inflect...



Source: AlphaWise, SIA, Morgan Stanley Research.

Exhibit 26 ...and power Analog unit momentum remains strong despite softer ASPs



Source: SIA, Morgan Stanley Research.

Raising Estimates for TXN

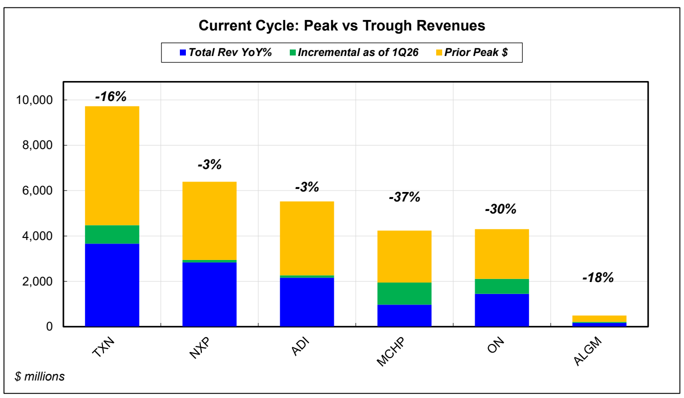
We believe the upcoming JunQ will be a different story than last year. Recall that JunQ 2025 appeared to carry similar tailwinds, with the broader analog complex speaking to clearly positive demand signals intra-quarter - only for the momentum to prove short-lived and for TXN to miss expectations. What we see now is different in character - yes, there is some sequential easing at the margin, but the group remains ~10% below historical peak levels overall, with Industrial similarly ~10% below peak and Auto holding at ~14% below peak. Data center provides additional support, with TXN benefiting from broad content across power mgmt, signal chain, clocking, sensing, protection and control used across compute, networking and rack power/cooling. We view this as additive to the Industrial recovery, rather than the sole reason for the estimate increase.

The m/m unevenness is not a new development. As the SIA data has consistently shown, m/m oscillations across analog/MCU have been highly uneven throughout this recovery, and we do not interpret the current normalization as an overall cooling of demand. What is new and notable is the breadth of the Industrial recovery - 1Q26 marked the first quarter where the broad market (including traditional factory automation, medical, and building automation) began to wake up alongside the more defensive verticals like A&D and energy that had been carrying the early recovery. For TI specifically, Industrial grew across all sectors and all geographies sequentially last quarter - a qualitatively different signal than what we saw in 1H25.

Why are we not raising estimates elsewhere? TXN is the first upcoming print and provides the earliest test of whether the survey's Analog/Industrial strength, data center momentum and firm pricing are translating into revenue and gross margin. The read-through for the rest of the group remains constructive but less immediate: Auto remains mixed, power tightness is selective and most MCHP pricing actions do not take effect until September, with the P&L impact weighted toward 4Q26. We prefer to use TXN's results and subsequent company commentary to confirm the breadth and durability of these trends before changing the remaining models.

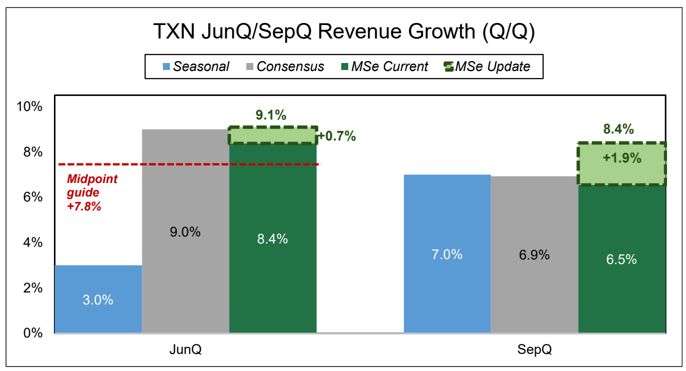
Changes to our estimates: We raise our JunQ/SepQ revenue growth estimates by 70bps/190bps to 9.1%/8.4% q/q, respectively, versus normal seasonality of ~3%/~7%. CY26 revenue/GPM/EPS move from $21.0bn/60.1%/$7.82 to $21.3bn/59.6%/$7.90. Our CY27 estimates move from $22.3bn/60.7%/$8.50 to $22.8bn/60.9%/$8.86. Our PT to moves up to $230 from $221, based on 26x CY27E EPS of $8.86. We are 1% above/3% below Street on revenue and 2% above/2% below Street on EPS for CY26/CY27, respectively.

Exhibit 27: Most broad analog names remain below prior peaks



Source: Company Data, Morgan Stanley Research.

Exhibit 28: We raise our JunQ/SepQ estimates for TXN

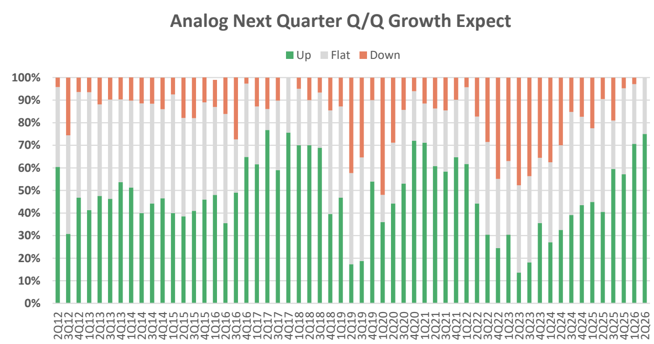


Source: Morgan Stanley Research Estimates, VisibleAlpha, Company Data.

2Q26 Results – Key Charts & Takeaways

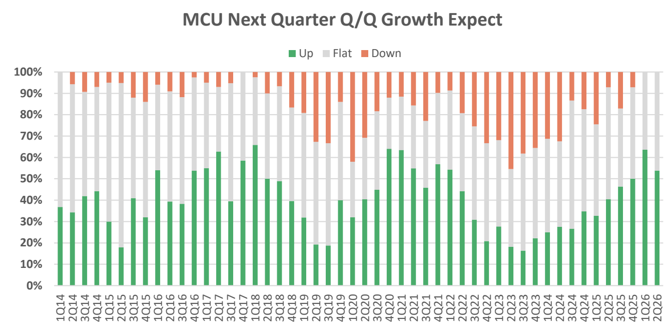
Main takeaway: Strong JunQ prints, with healthy SepQ expectations across Analog/MCU

Exhibit 29: Analog growth expectations increased from 1Q...



Source: AlphaWise, Morgan Stanley Research

Exhibit 30: ...while MCU growth expectations eased marginally, though also remain healthy

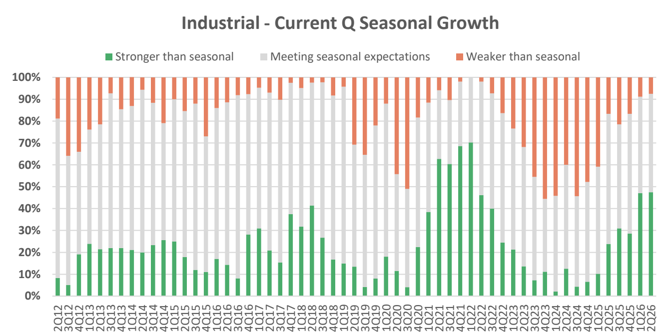


Source: AlphaWise, Morgan Stanley Research

1) Industrial Holds Up; Auto Remains Mixed

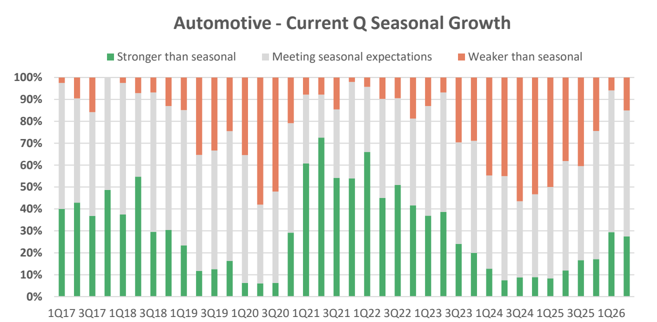
Industrial momentum held up, with stronger-than-seasonal expectations remaining ~flat q/q. The latest Multi-Industry survey was also supportive, while company commentary continues to point to underlying demand rather than inventory build. Auto remains mixed, with weaker-than-seasonal expectations increasing even as inventory digestion appears largely complete.

Exhibit 31: Stronger-than-seasonal Industrial expectations remained ~flat q/q



Source: AlphaWise, Morgan Stanley Research

Exhibit 32: ...while weaker-than-seasonal Auto expectations increased to 15% from 6%

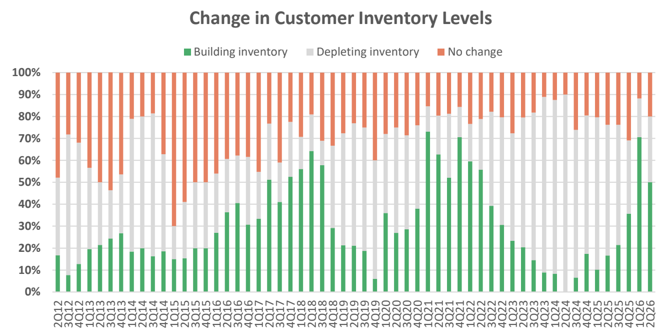


Source: AlphaWise, Morgan Stanley Research

2) Inventory Builds Moderate After the 1Q Jump

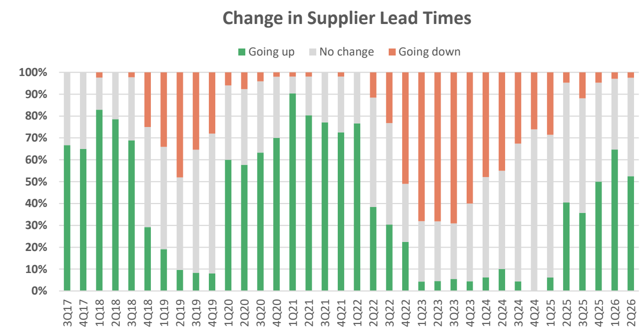
Overall inventory-building intentions declined to 50% from 71%, while depletion increased to 30% from 18%. However, builds greater than 5% remained ~flat for Analog and MCU. Combined with distributor inventory declines and below-seasonal producer builds, the data point to continued inventory discipline despite elevated absolute levels, rather than broad restocking or renewed digestion.

Exhibit 33: Those looking to deplete inventory increased from 18% in the MarQ to 30% in the JunQ...



Source: AlphaWise, Morgan Stanley Research

Exhibit 34: ... expectations for increasing supplier lead times decreased from 65% to 53%...

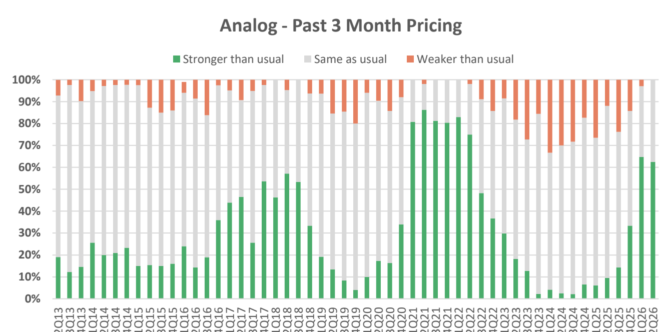


Source: AlphaWise, Morgan Stanley Research

3) Pricing Holds; Tightness Remains Selective

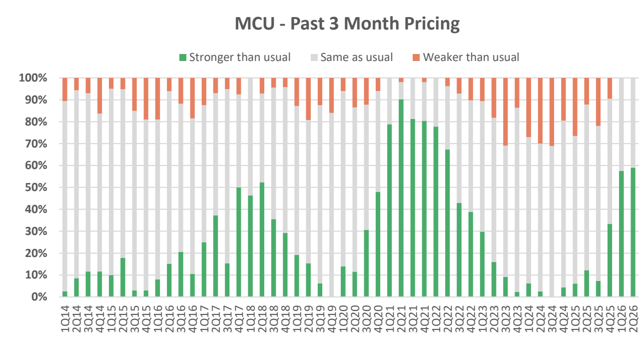
Pricing remained firm, with 63% reporting stronger-than-usual Analog pricing and 59% reporting stronger MCU pricing, while weaker-pricing responses fell to zero. Overall constraints remained unchanged and lead-time momentum eased, but Analog expedites, MCU constraints and MCU double ordering increased. The data support firm pricing and selected areas of tightening, rather than a broad shortage call.

Exhibit 35: Stronger-than-usual Analog pricing held near the 1Q26 high (63% vs 65%)...



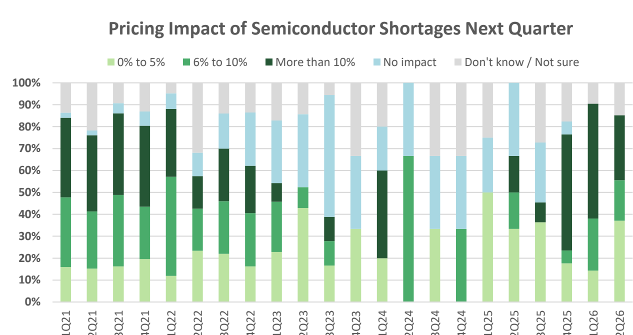
Source: AlphaWise, Morgan Stanley Research

Exhibit 36: ...while stronger MCU pricing remained ~flat q/q...



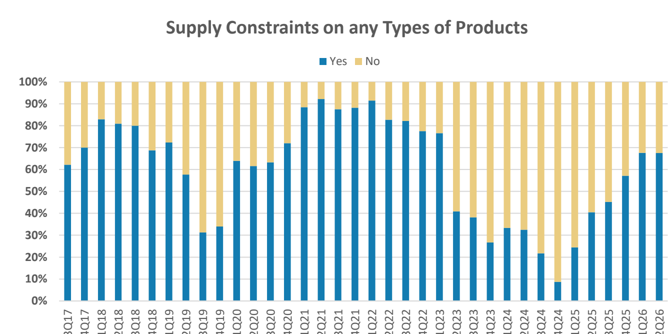
Source: AlphaWise, Morgan Stanley Research

Exhibit 37: ...and the magnitude of pricing pressure remained elevated...



Source: AlphaWise, Morgan Stanley Research

Exhibit 38: with overall supply constraints holding at 68%



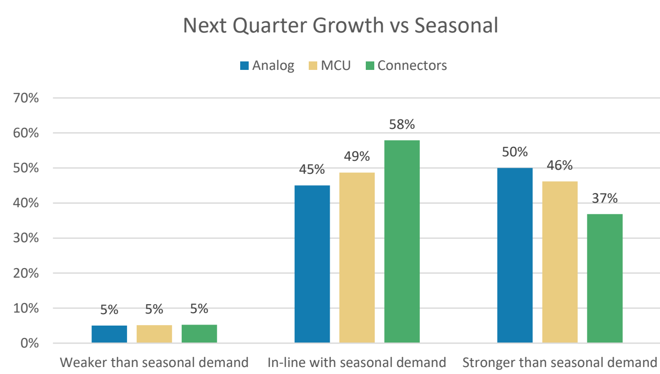
Source: AlphaWise, Morgan Stanley Research

2Q'26 Disti Survey - Details

In conjunction with Morgan Stanley AlphaWise, we conducted our 55th quarterly survey among 40 semiconductor distributors during May - July 2026. We asked questions around sales volumes, growth expectations and inventory levels, among other topics.

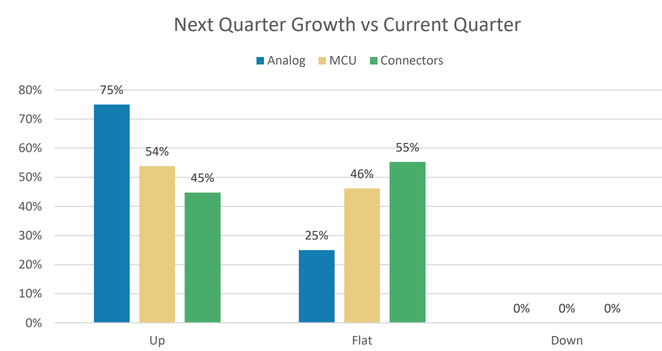
SepQ expectations remain healthy across Analog/MCU after accounting for seasonality

Exhibit 39: The majority of respondents anticipate results in-line with or stronger than seasonal...



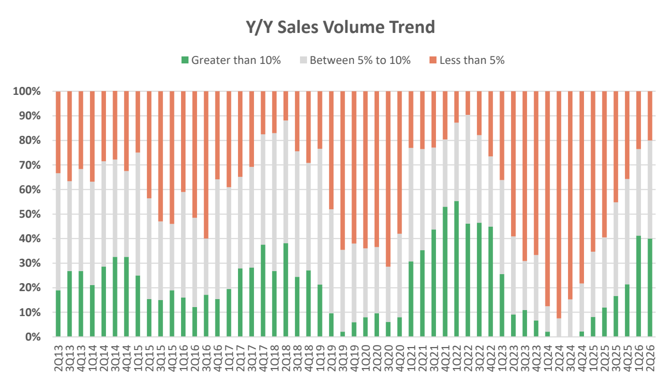
Source: AlphaWise, Morgan Stanley Research

Exhibit 40: ...while sequential 3Q26 growth expectations remain healthy.



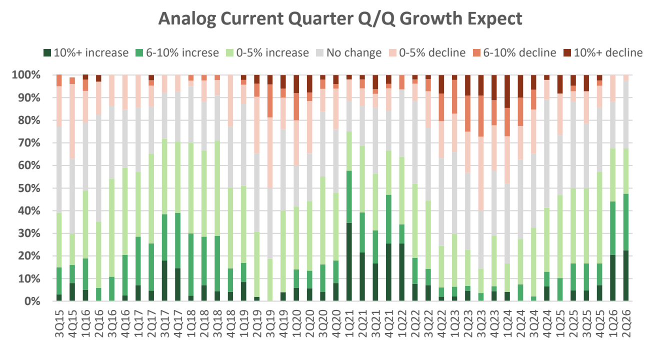
Source: AlphaWise, Morgan Stanley Research

Exhibit 41: Respondents tracking at less than 5% y/y sales growth declined further to 20% from 24% last quarter



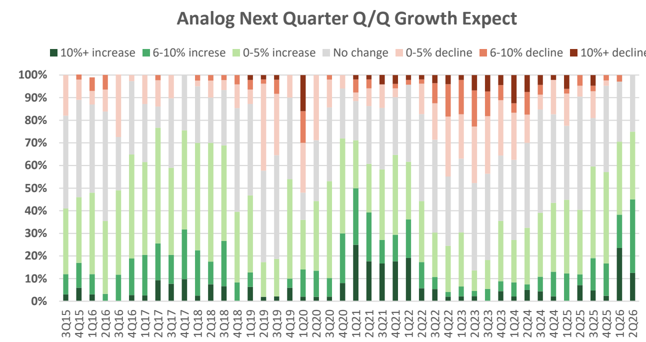
Source: AlphaWise, Morgan Stanley Research

Exhibit 42: 3% of respondents expect a sequential decline for analog in June, down from 12% last quarter…



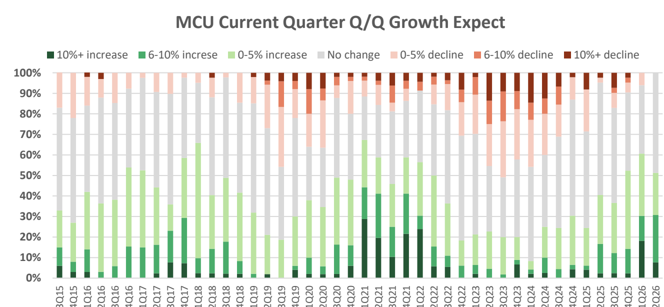
Source: AlphaWise, Morgan Stanley Research

Exhibit 43: …while none expect a sequential decline in the September quarter, down from 3% going into the June quarter



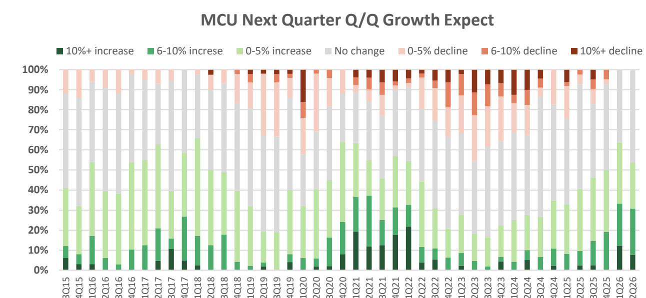
Source: AlphaWise, Morgan Stanley Research

Exhibit 44: No respondents expect a sequential decline for MCU…



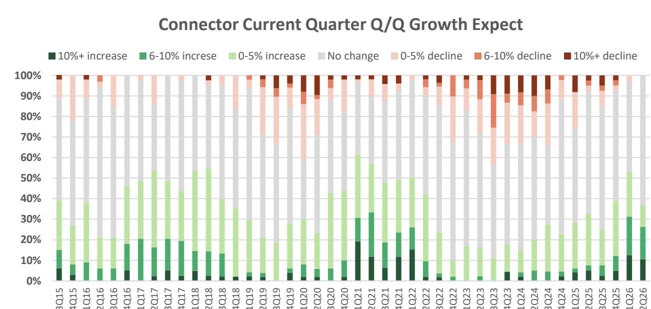
Source: AlphaWise, Morgan Stanley Research

Exhibit 45: …while 54% of respondents expect sequential growth in the September quarter, down from 64% going into the June quarter



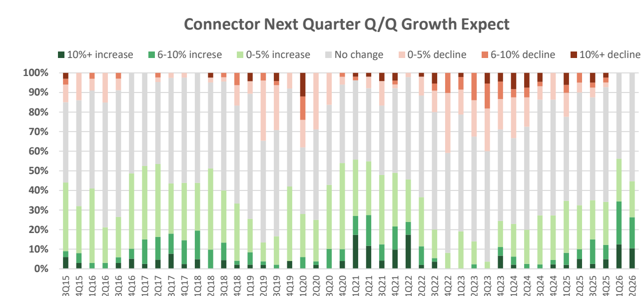
Source: AlphaWise, Morgan Stanley Research

Exhibit 46: No respondents expect a sequential decline for connectors in June, down from 3% last quarter…



Source: AlphaWise, Morgan Stanley Research

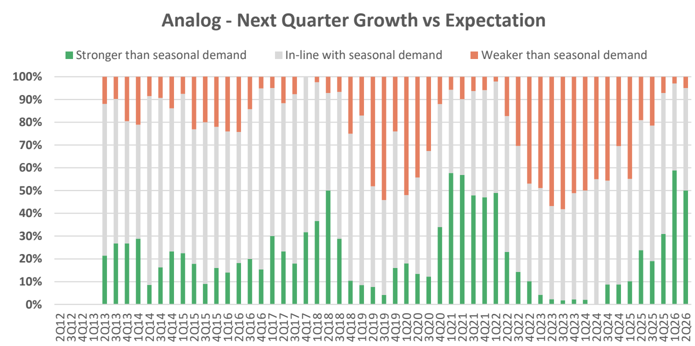
Exhibit 47: …while 45% of respondents expect sequential growth in the September quarter, down from 56% going into the June quarter



Source: AlphaWise, Morgan Stanley Research

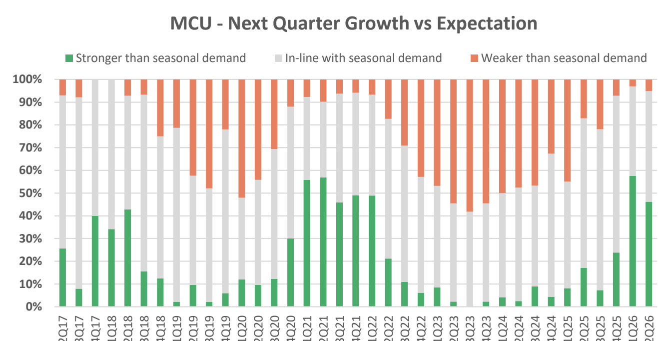
Above-seasonal Analog/MCU expectations eased from 1Q, but remain elevated

Exhibit 48: 50% of respondents expect a stronger-than-seasonal September quarter for analog, down from 59% last quarter...



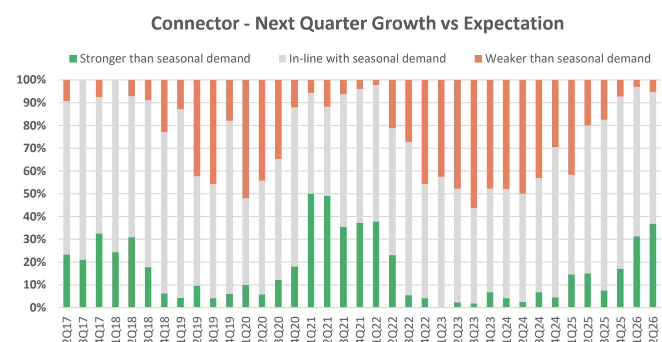
Source: AlphaWise, Morgan Stanley Research

Exhibit 49: ... while 46% of respondents expect a stronger-than-seasonal September quarter for MCU, down from 58% last quarter...



Source: AlphaWise, Morgan Stanley Research

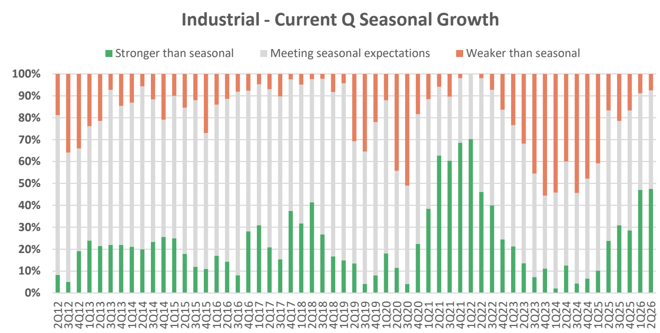
Exhibit 50: ... and 37% of respondents expect a stronger-than-seasonal September quarter for connectors, up from 31% last quarter



Source: AlphaWise, Morgan Stanley Research

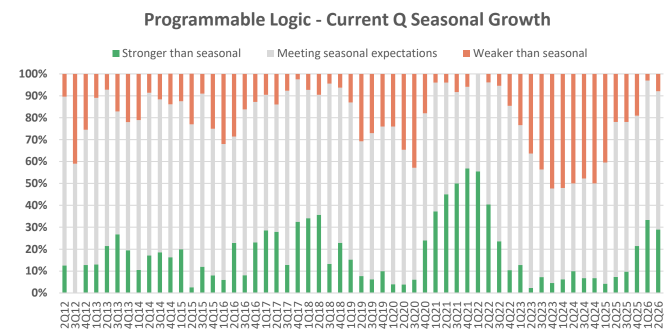
End-market trends were mixed, with Industrial holding and Auto softening

Exhibit 51: 47% expect stronger-than-seasonal industrial sales, ~flat from last quarter



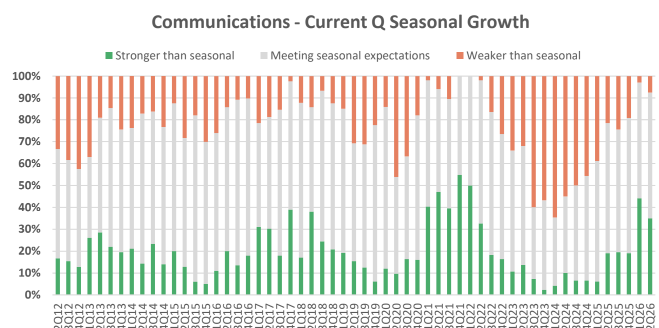
Source: AlphaWise, Morgan Stanley Research

Exhibit 52: 29% expect stronger-than-seasonal PLD sales, down from 33% last quarter



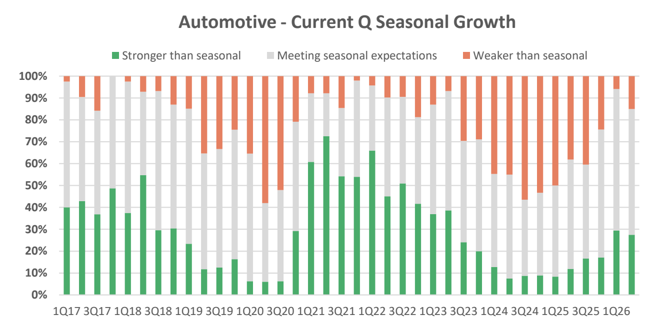
Source: AlphaWise, Morgan Stanley Research

Exhibit 53: 35% expect weaker-than-seasonal communications sales, down from 44% last quarter



Source: AlphaWise, Morgan Stanley Research

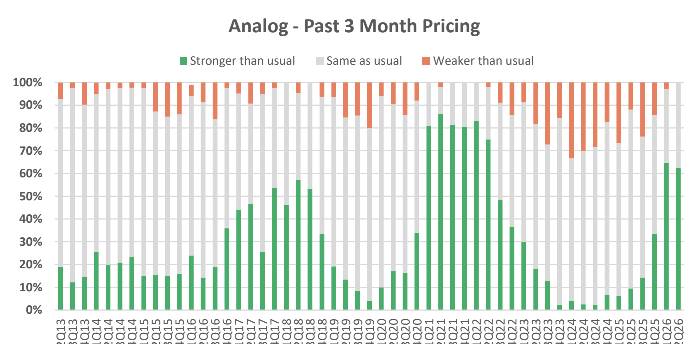
Exhibit 54: 15% expect weaker-than-seasonal automotive sales, up from 6% last quarter



Source: AlphaWise, Morgan Stanley Research

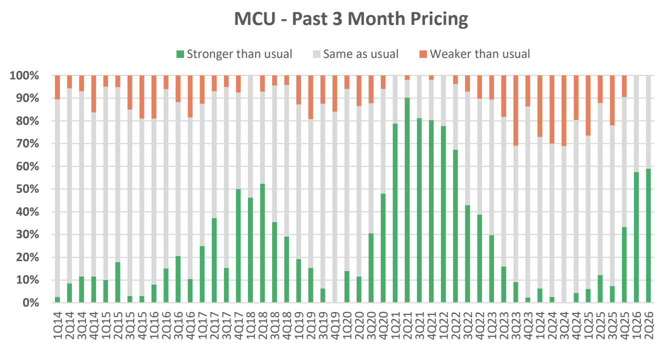
Stronger pricing broadly held firm across product lines

Exhibit 55: Respondents seeing stronger-than-usual analog pricing was ~flat at 63% from 65%...



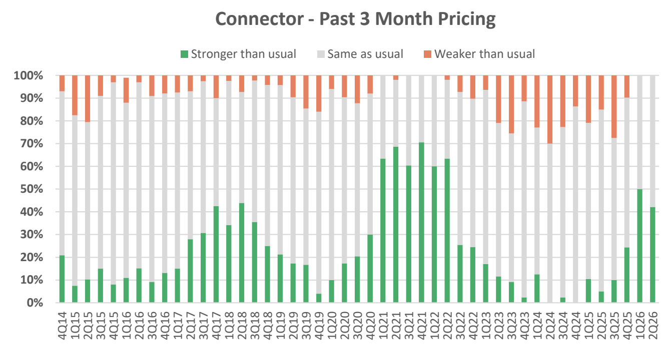
Source: AlphaWise, Morgan Stanley Research

Exhibit 56: … and 59% of respondents saw stronger MCU pricing, ~flat from 58% last quarter…



Source: AlphaWise, Morgan Stanley Research

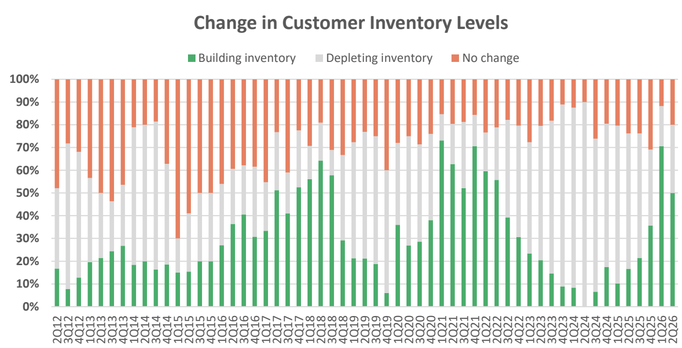
Exhibit 57: … and no respondents saw weaker connector pricing, in-line with last quarter



Source: AlphaWise, Morgan Stanley Research

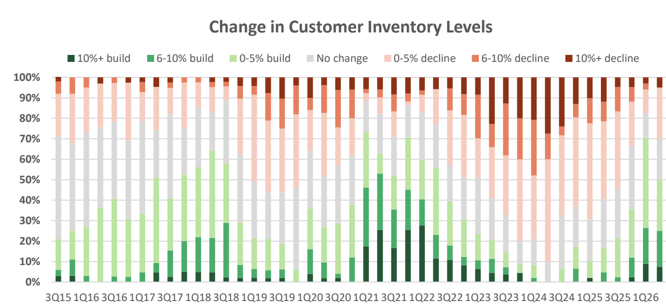
Across product lines, inventory builds moderated q/q, while depletion increased

Exhibit 58: 30% of respondents are looking to deplete inventory, up from 18% last quarter



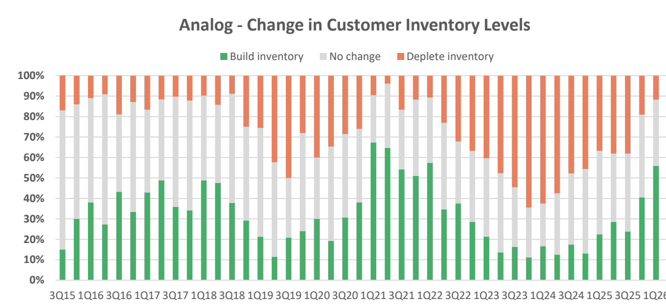
Source: AlphaWise, Morgan Stanley Research

Exhibit 59: 5% of respondents are looking to deplete inventory by greater than 5%, ~flat vs 6% last quarter



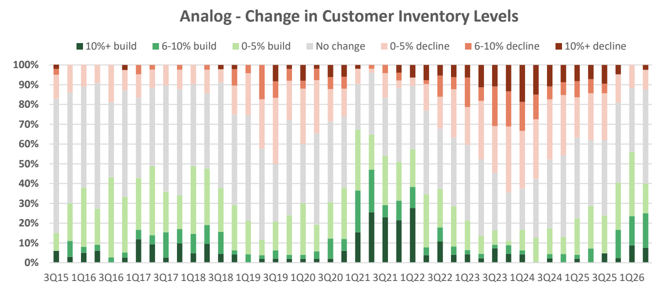
Source: AlphaWise, Morgan Stanley Research

Exhibit 60: In analog, 13% of respondents are looking to deplete inventory, ~flat vs 12% last quarter…



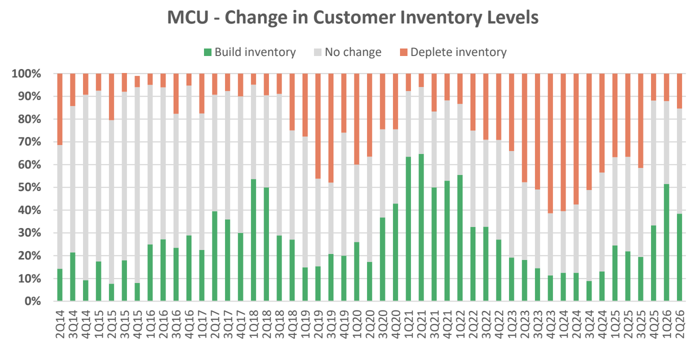
Source: AlphaWise, Morgan Stanley Research

Exhibit 61: ... and 25% are looking to build inventory by more than 5%, ~flat vs 24% last quarter



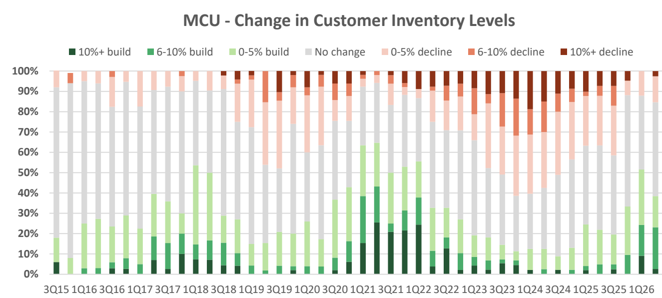
Source: AlphaWise, Morgan Stanley Research

Exhibit 62: In MCUs, 38% of respondents are looking to build inventory, down from 52% last quarter...



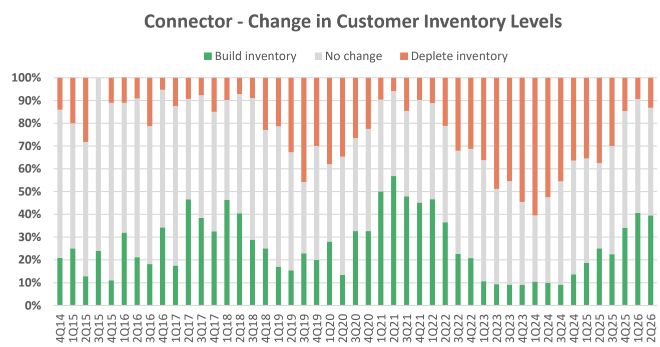
Source: AlphaWise, Morgan Stanley Research

Exhibit 63: ... and 23% are looking to build by more than 5%, ~flat vs 24% last quarter



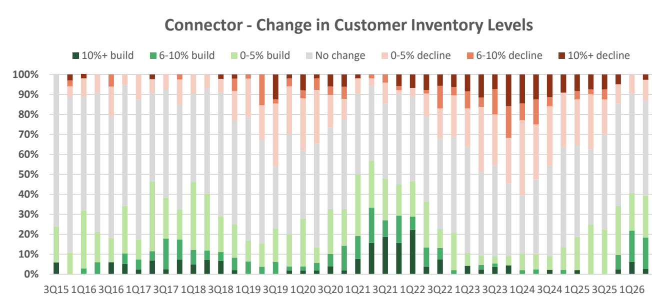
Source: AlphaWise, Morgan Stanley Research

Exhibit 64: In connectors, 13% of respondents are looking to deplete inventory, up from 9% last quarter...



Source: AlphaWise, Morgan Stanley Research

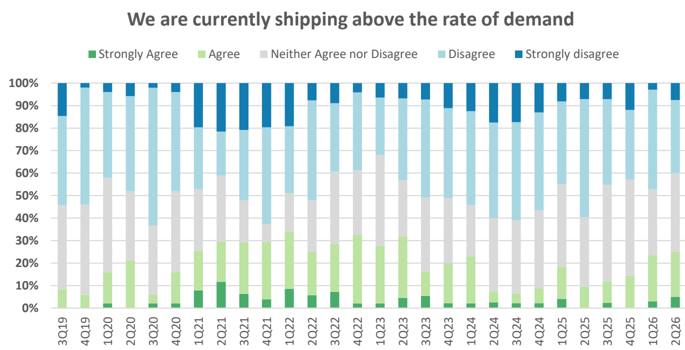
Exhibit 65: ... and 18% are looking to build by more than 5%, down from 22% last quarter



Source: AlphaWise, Morgan Stanley Research

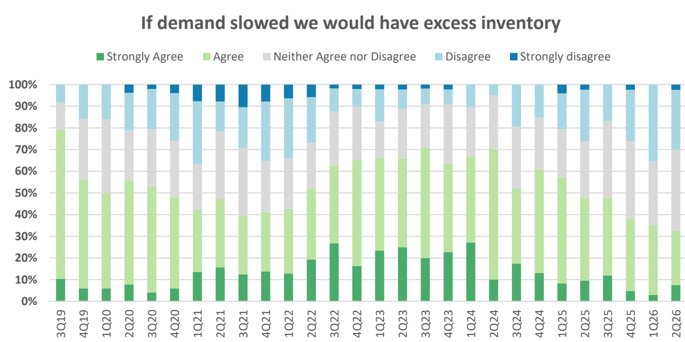
Most respondents remain at or below the rate of demand, while concerns around supply availability increased

Exhibit 66: 20% of respondents believe they are shipping above the rate of demand, ~flat from 21% last quarter



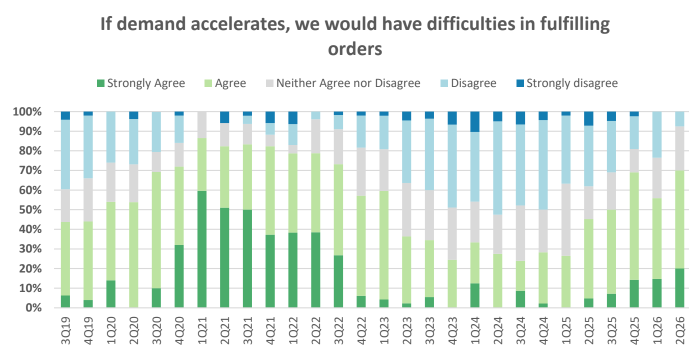
Source: AlphaWise, Morgan Stanley Research

Exhibit 67: 25% of respondents believe that if demand slowed they would be holding excess inventory, down from 32% last quarter



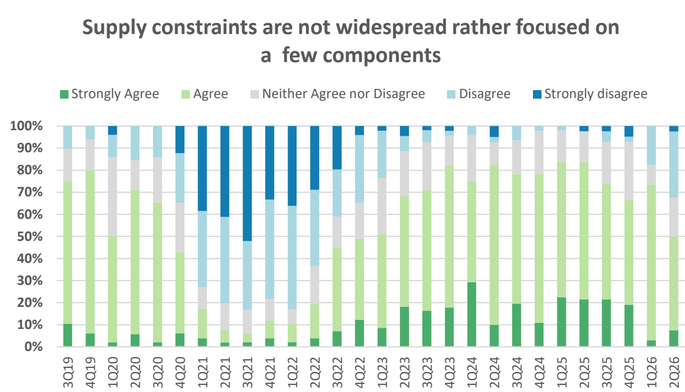
Source: AlphaWise, Morgan Stanley Research

Exhibit 68: 50% of respondents believe that they would have difficulty fulfilling orders if demand accelerates, up from 41% last quarter



Source: AlphaWise, Morgan Stanley Research

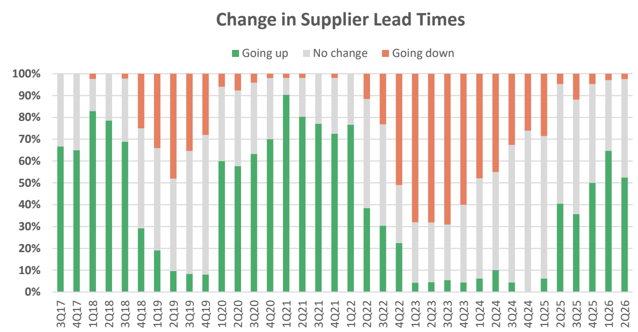
Exhibit 69: 43% of respondents believe that supply constraints are not widespread, down from 71% last quarter



Source: AlphaWise, Morgan Stanley Research

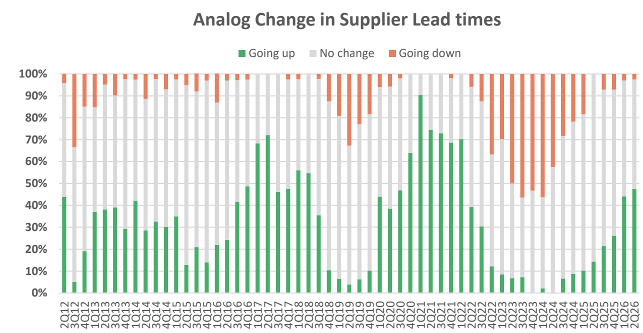
Lead-time increases remain elevated, but broader momentum eased

Exhibit 70: 53% of respondents reported that supplier lead times are going up, down from 65% last quarter



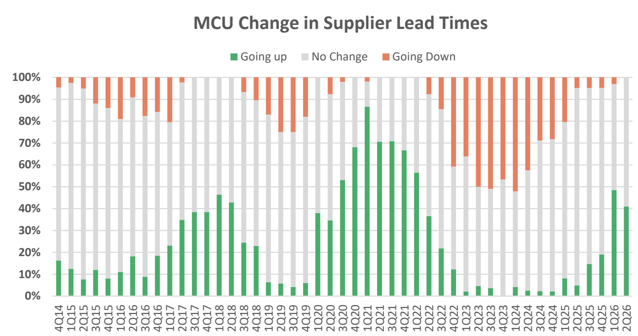
Source: AlphaWise, Morgan Stanley Research

Exhibit 71: 3% of respondents reported analog lead times are decreasing, in-line with last quarter



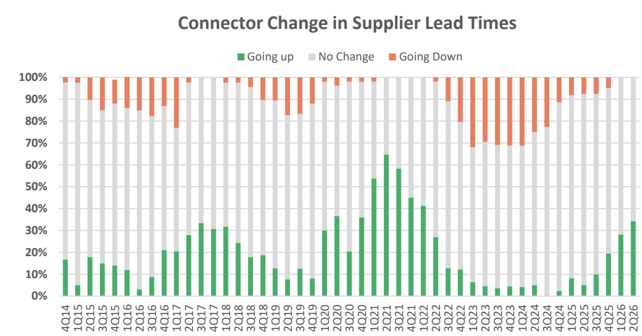
Source: AlphaWise, Morgan Stanley Research

Exhibit 72: 41% of respondents reported MCU lead times are increasing, down from 48% last quarter



Source: AlphaWise, Morgan Stanley Research

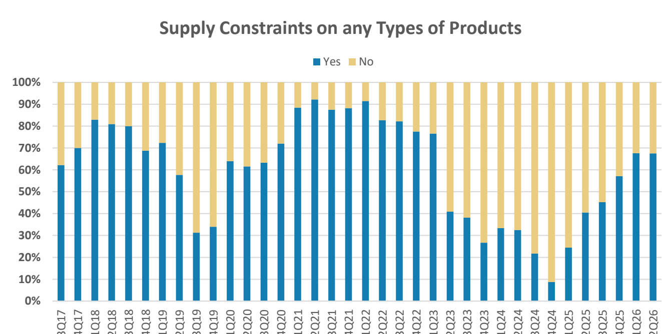
Exhibit 73: 34% of respondents reported connector lead times are increasing, up from 28% last quarter



Source: AlphaWise, Morgan Stanley Research

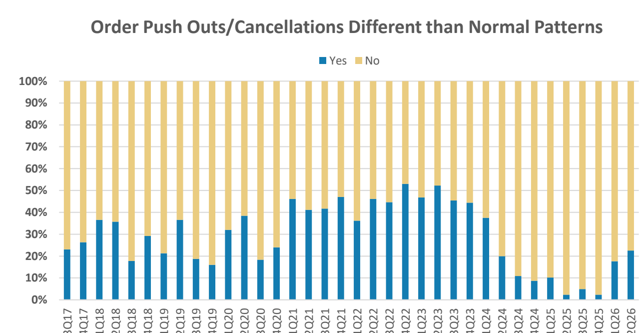
Analog constraints eased while MCU constraints increased; memory remained the most frequently cited constraint, with other responses citing power, Industrial and Auto MCUs

Exhibit 74: Respondents reporting supply constraints on at least one product remained in-line with last quarter at 68%



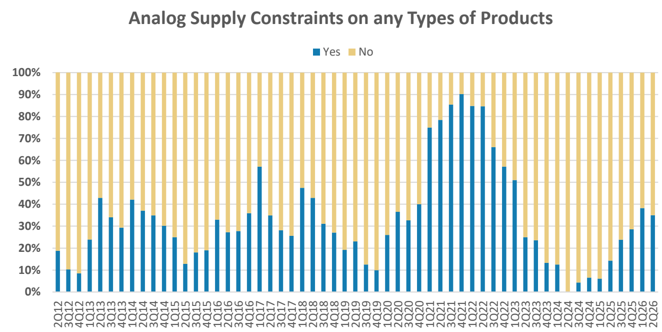
Source: AlphaWise, Morgan Stanley Research

Exhibit 75: Respondents reporting abnormal order push-outs or cancellations on at least one product increased from 18% to 23%



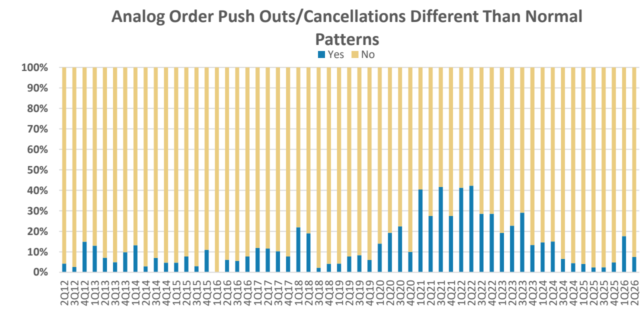
Source: AlphaWise, Morgan Stanley Research

Exhibit 76: Analog supply constraints decreased marginally from 38% to 35%



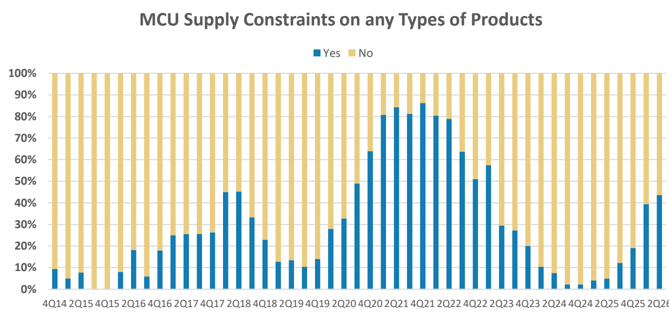
Source: AlphaWise, Morgan Stanley Research

Exhibit 77: Analog order push-outs and cancellations decreased from 18% to 8%



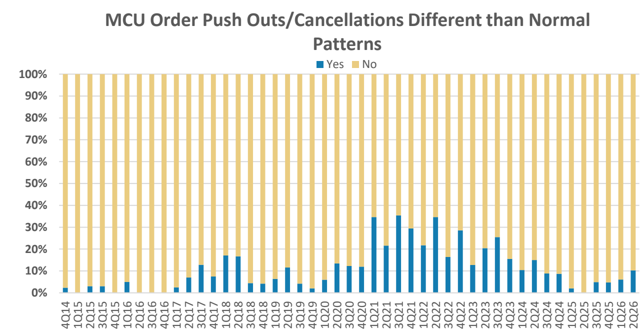
Source: AlphaWise, Morgan Stanley Research

Exhibit 78: MCU supply constraints increased from 39% to 44%



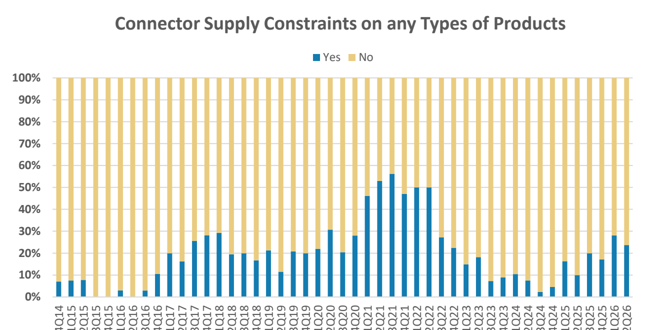
Source: AlphaWise, Morgan Stanley Research

Exhibit 79: MCU push-outs and cancellations increased from 6% to 10%



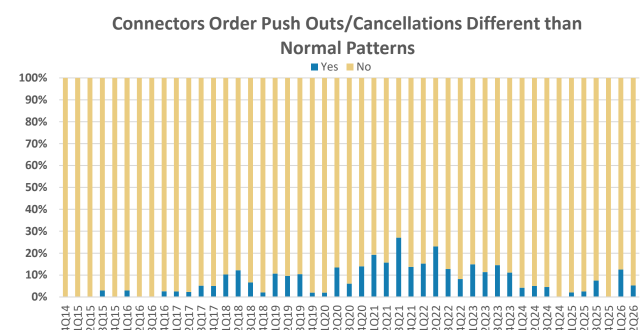
Source: AlphaWise, Morgan Stanley Research

Exhibit 80: Connector supply constraints decreased from 28% to 24%



Source: AlphaWise, Morgan Stanley Research

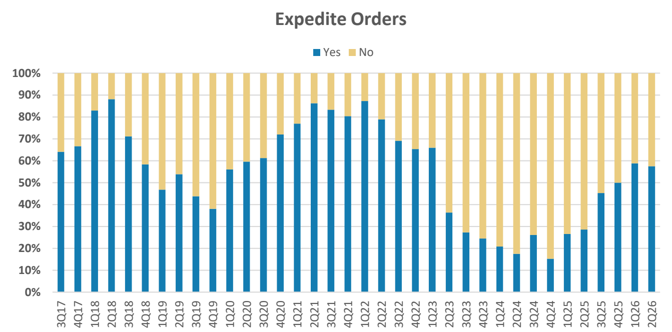
Exhibit 81: Connector push-outs and cancellations decreased from 13% to 5%



Source: AlphaWise, Morgan Stanley Research

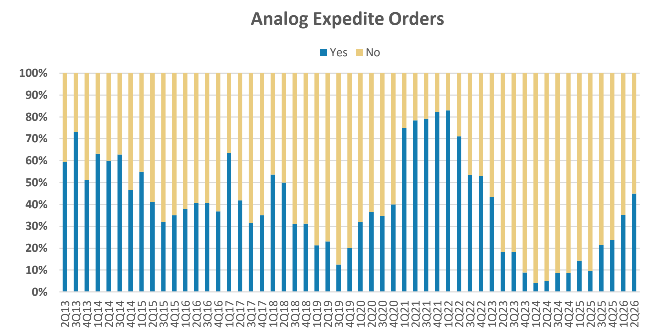
Expedites remained elevated, while MCU/connector double ordering increased

Exhibit 82: Expedite orders held ~flat at 58% vs 58% last quarter



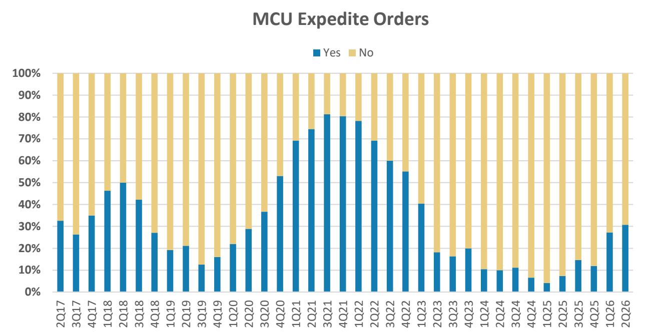
Source: AlphaWise, Morgan Stanley Research

Exhibit 83: Analog order expedites increased from 35% to 45%



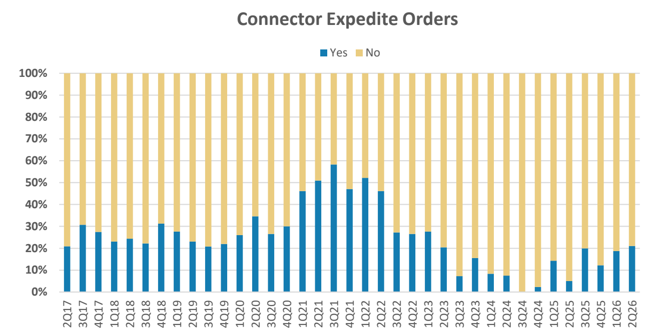
Source: AlphaWise, Morgan Stanley Research

Exhibit 84: MCU order expedites increased from 27% to 31%



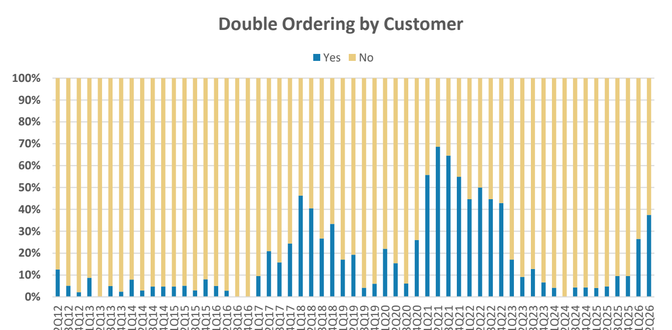
Source: AlphaWise, Morgan Stanley Research

Exhibit 85: Connector order expedites held ~flat at 21% vs 19%



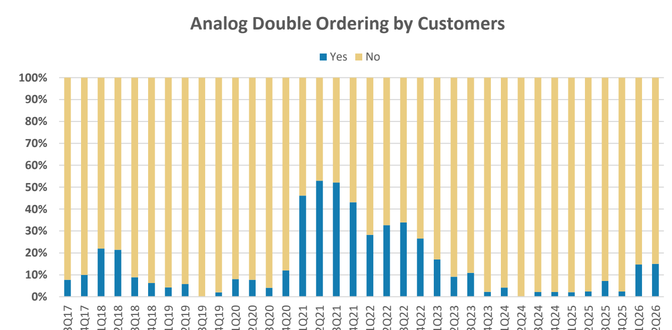
Source: AlphaWise, Morgan Stanley Research

Exhibit 86: Double ordering increased from 26% to 38%



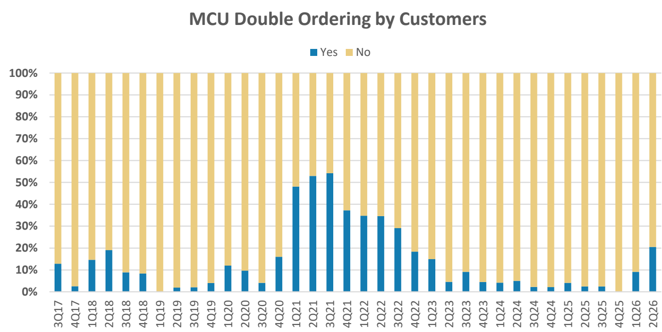
Source: AlphaWise, Morgan Stanley Research

Exhibit 87: Analog double ordering remained in-line with last quarter at 15%



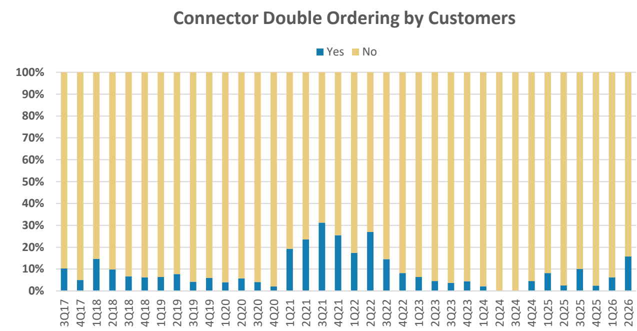
Source: AlphaWise, Morgan Stanley Research

Exhibit 88: MCU double ordering increased from 9% to 21%



Source: AlphaWise, Morgan Stanley Research

Exhibit 89: Connector double ordering increased from 6% to 16%

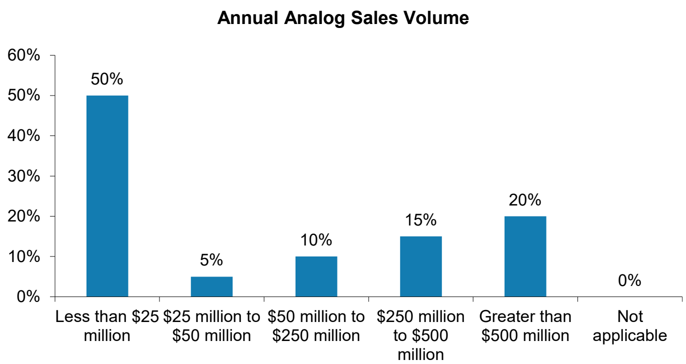


Source: AlphaWise, Morgan Stanley Research

Survey Sample Profile

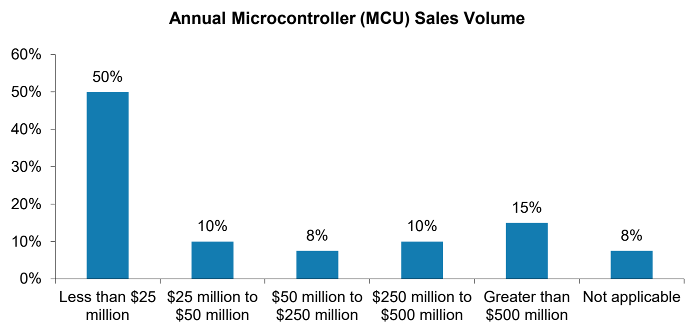
Morgan Stanley AlphaWise surveyed 40 respondents in the North American distribution channel to gauge market conditions. % of respondents with sales volume of over $250mn were as follows; Analog 35%, MCU 25%, PLD 17.5%, Discrete 20%, Connectors 20%.

Exhibit 90: Analog Survey Respondents



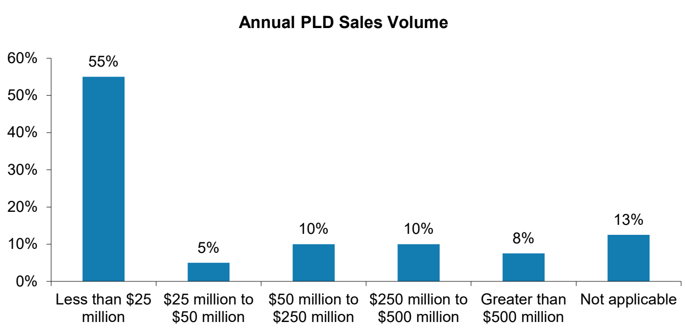
Source: AlphaWise, Morgan Stanley Research

Exhibit 91: MCU Survey Respondents



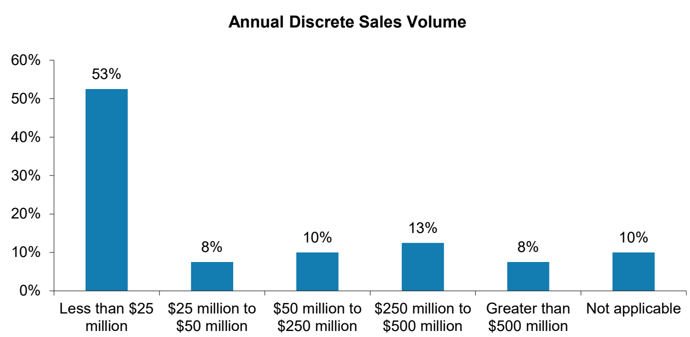
Source: AlphaWise, Morgan Stanley Research

Exhibit 92: PLD Survey Respondents



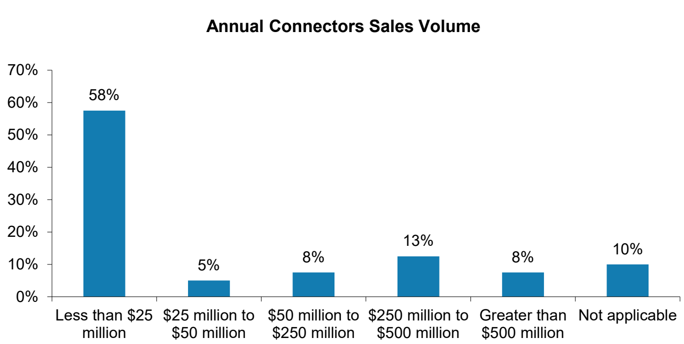
Source: AlphaWise, Morgan Stanley Research

Exhibit 93: Discrete Survey Respondents



Source: AlphaWise, Morgan Stanley Research

Exhibit 94: Connector Survey Respondents



Source: AlphaWise, Morgan Stanley Research

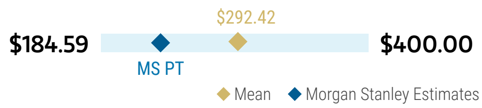
Risk Reward – Texas Instruments (TXN.O)

Underweight on limited upside given premium valuation & margin contraction

PRICE TARGET $230.00

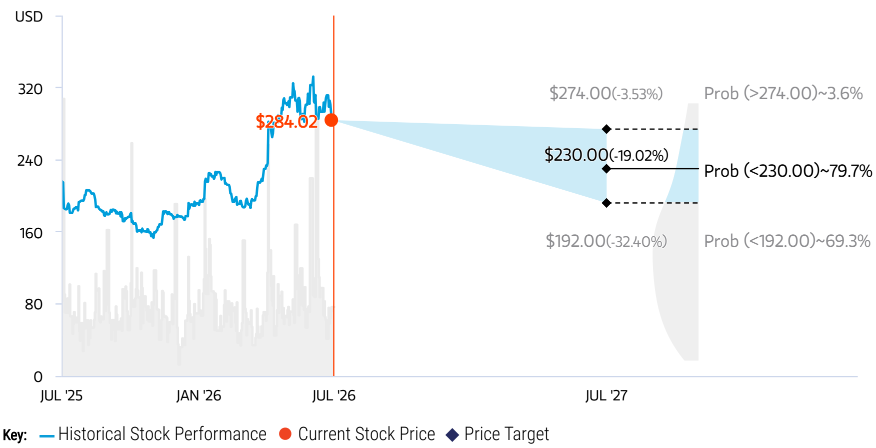
Represents 26x our CY27e EPS of $8.86, reflecting mid-cycle positioning, a slight premium to its long-term average of 24x and in-line with large-cap analog peers.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)

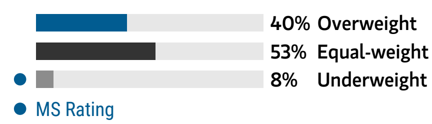


Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 17 Jul 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology here

UNDERWEIGHT THESIS

- TI is looking to ramp capex over the next few years to help bring more manufacturing in-house, but this will act as a near-term headwind to earnings and FCF growth. The company has also seen pricing and market share pressure in personal electronics and enterprise, as customers move to multisource components, and face competition with local Chinese supply.
- The company trades at a premium valuation for its solid governance track record, and we admire the company's long-term direction, but we see this dynamic as limiting upside in the near-term as margin should come under pressure from the capex ramp.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Negative*

View descriptions of Risk Rewards Themes here

BULL CASE $274.00

28x Bull Case 2027e EPS of $9.78

Rebound in macroeconomic environment drives analog and embedded snapback, and end markets grow more than expected in 2026 & 2027.

- Revenue growth 13.5% in 2027

-Gross margin expands to 62.4% in 2027

- EPS of $9.78 in 2027

BASE CASE $230.00

26x Base case 2027e EPS of $8.86

Assumes a moderate analog and embedded recovery through 2026 & 2027.

- 20.7% revenue growth in 2026 and 7.1% revenue growth in 2027

- TXN achieves 59.6% GMs in 2026 and 60.9% in 2027

- EPS of $7.90 in CY26 and $8.86 in CY27

BEAR CASE $192.00

24x Bear Case 2027e EPS of $7.98

Global macroeconomic weakness slows down the recovery in analog and embedded, and end markets remain weak.

- 0.7% revenue growth in 2027

- Gross margin of 59.4% in 20267

- EPS of $7.98 in 2027

Risk Reward – Texas Instruments (TXN.O)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
GAAP Revenue ($, mm)	17,682	21,334	22,848	24,123
MW Gross Margin (%)	57.0	59.6	60.9	61.9
GAAP EPS ($)	5.45	7.90	8.86	9.79
Inventory ($, mm)	4,804	4,698	4,422	4,188
DOI	227.6	196.2	178.3	164.0

INVESTMENT DRIVERS

- Analog and embedded growth vs industry peers (ADI for analog, STM/Infineon/Renesas/NXP/MCHP for embedded).
- How increased capex will impact revenue growth and market share as well as depreciation burden on gross margin.

GLOBAL REVENUE EXPOSURE

- 0-10% APAC, ex Japan, Mainland China and India
- 0-10% Japan
- 0-10% UK
- 10-20% Mainland China
- 20-30% Europe ex UK
- 30-40% North America

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 BEST	24 Month Horizon	5/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Increasing domestic capacity leads to outsized market share gains.
- Recovery in consumer markets (personal electronics and embedded).
- Continued content gains in Auto and Industrial end-markets.

RISKS TO DOWNSIDE

- Continued market share loss in the embedded business.
- Pricing and market share pressure from China local capacity.
- Market share loss to ADI in the analog business.

OWNERSHIP POSITIONING

Inst. Owners, % Active	49.7%
HF Sector Long/Short Ratio	2.1x
HF Sector Net Exposure	29.5%

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue ($, mm)	19,666	21,334	21,043	22,034
EBITDA ($, mm)	9,447	10,999	10,723	11,808
Net income ($, mm)	5,781	7,195	6,998	7,638
EPS ($)	7.16	7.90	7.77	8.32

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of June 30, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

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Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

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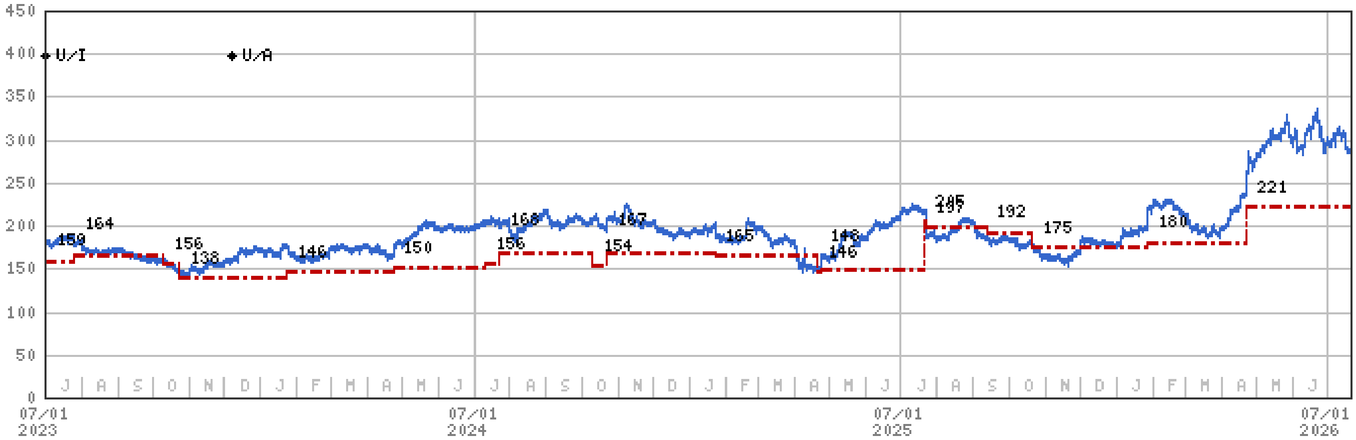
Stock Price, Price Target and Rating History (See Rating Definitions)

Texas Instruments (TXN.O) - As of 07/20/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 7/1/21 : U/I; 12/7/23 : U/A

Price Target History: 4/12/21 : 176; 7/19/21 : 185; 7/22/21 : 193; 10/26/21 : 195; 1/25/22 : 198; 4/25/22 : 181; 4/26/22 : 170; 6/9/22 : 155; 7/26/22 : 160; 10/25/22 : 152; 1/24/23 : 166; 4/25/23 : 159; 7/25/23 : 164; 10/10/23 : 156; 10/24/23 : 138; 1/23/24 : 146; 4/23/24 : 150; 7/10/24 : 156; 7/23/24 : 168; 10/10/24 : 154; 10/22/24 : 167; 1/23/25 : 165; 4/20/25 : 146; 4/23/25 : 148; 7/21/25 : 205; 7/22/25 : 197; 9/12/25 : 192; 10/21/25 : 175; 1/27/26 : 180; 4/22/26 : 221

Source: Morgan Stanley Research   Date Format : MM/DD/YY   Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst)   Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/17/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	$495.76
Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	$16.71
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	$46.48
Ambarella Inc (AMBA.O)	O (03/29/2016)	$61.84
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	$62.94
Analog Devices Inc. (ADI.O)	O (11/16/2023)	$375.36
Astera Labs Inc (ALAB.O)	O (05/11/2025)	$303.62
Broadcom Inc. (AVGO.O)	O (06/09/2024)	$370.83
Cerebras Systems (CBRS.O)	O (06/08/2026)	$172.86
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	$57.48
Intel Corporation (INTC.O)	E (02/22/2023)	$95.04
IonQ Inc (IONQ.N)	E (04/25/2023)	$34.78
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	$188.68
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	$80.96
Micron Technology Inc. (MU.O)	O (10/06/2025)	$848.95
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	$11.46
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	$202.81
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	$266.53
ON Semiconductor Corp. (ON.O)	++	$87.37
Qorvo Inc (QRVO.O)	E (10/28/2025)	$85.50
Qualcomm Inc. (QCOM.O)	E (06/24/2026)	$171.78
Quantinuum (QNT.O)	E (06/29/2026)	$56.07
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	$1,354.82
Semtech Corp. (SMTC.O)	E (04/06/2025)	$124.96
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	$217.47
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	$59.35
Texas Instruments (TXN.O)	U (04/13/2020)	$284.02

Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	$29.89
Lee Simpson		
Arm Holdings plc (ARM.O)	E (04/07/2026)	$267.19
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	$330.11
Synopsys Inc. (SNPS.O)	E (02/27/2026)	$384.28

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.